A SEARCH FOR TIME-VARIABLE AND FREQUENCY-DRIFTING NARROWBAND TECHNOSIGNATURES TOWARD ALMA-OBSERVED STARS WITHIN 40 PC OF THE SUN: SURVEY DESIGN AND FIRST RESULTS

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ABSTRACT

We have searched public archival ALMA observations of a volume-limited census of 168 stars within 40 pc of the Sun for narrowband technosignatures. We find no evidence for a technosignature in any of the data searched. The experiment constrains carriers that are time-variable on the timescale of an observing track and/or drift in frequency: the per-channel temporal-median subtraction that rejects stationary ground-based interference also suppresses phase-steady carriers, so a stable beacon lies outside the class this pipeline can constrain, and we make no statement about it. At the epoch of this release, 104 of 208 planned star/band datasets across 85 census stars (79 independent systems) have been searched to completion: 417 spectral windows spanning 1.3–38.8 pc and 92.4 GHz of unique sky frequency between 89.6 and 495.1 GHz. Every window's stellar position is compared with 512 control positions carried through an identical search, which calibrates the false-alarm rate empirically rather than assuming Gaussian statistics. Four windows pass the joint criterion: three are circumstellar CO (two toward β Pictoris, one toward HD 48370) and one, toward CP$-$72 2713, is a marginal crossing consistent with the 0.8 flags expected by chance across 417 windows.

Nominal 5σ detection thresholds span 1.6×10^{13}–2.8×10^{17} W in EIRP (median 1.3×10^{15} W). Because most windows are channelised at 15.625 MHz these are native-channel power thresholds, not Hz-resolution narrowband sensitivities. A 1200-trial injection campaign into real calibrated visibility data recovers the searched class at 42^{+16}_{-14} per cent at the nominal threshold, so these thresholds must not be read as 95-per-cent completeness limits. Folding the measured completeness in, and conditioning explicitly on emission lying inside the searched spectral windows and persisting through the observed epochs, the zero-detection result bounds the fraction of systems hosting a transmitter of EIRP $\geq 10^{16}$ W at < 6.5 per cent (95 per cent confidence; the 56 systems with injection-calibrated coverage). Idealised calculations assuming unit recovery, or a uniform 42-per-cent recovery, would instead give 3.8 and 9.0 per cent; we quote them only as bracketing values. All limits assume continuous emission: at duty cycle 0.5 the primary bound weakens to 13 per cent, and by 0.1 it is 65 per cent and effectively vacuous.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, whether deliberately signalling or simply going about its own affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches have grown into surveys of many thousands of stars, yet, as Wright et al. (2018) quantified, they have sampled only an infinitesimal fraction of the multi-dimensional space of transmitter frequency, bandwidth, power, and duty cycle. Three gaps persist: the millimetre and submillimetre regime above ~50 GHz is essentially unsearched; most surveys target interest-selected classes of star rather than the local stellar population; and few studies have systematically reprocessed *existing* interferometric archives.

This paper addresses all three. We construct a census of every star within 40 pc of the Sun with public ALMA archival coverage, irrespective of spectral type, disc-hosting status, or known planets, and mine that archive for narrowband technosignature candidates and anomalous millimetre continuum, correcting for stellar proper motion and for the Doppler drift rates plausible for a transmitter on an orbiting platform. The search class is set by the processing itself, which suppresses phase-steady carriers, so this paper constrains *time-variable and/or sufficiently frequency-drifting narrowband emission* only (Appendix E measures both sides of that definition). Four archival-specific analyses supplement the classical search: closure-phase vetting, a continuum anomaly screen, a chirp and periodicity re-analysis, and a cross-target frequency-occupancy check.

Per-target tables are deferred to online-only Appendix I. We publish at 41 per cent completion because the citable content is methodological – the census definition, the empirically calibrated false-alarm machinery, and the measured completeness – together with a first set of per-target limits, every number tied to a frozen pipeline and a named data snapshot that later work will supersede row by row without invalidating this one.

2. BACKGROUND

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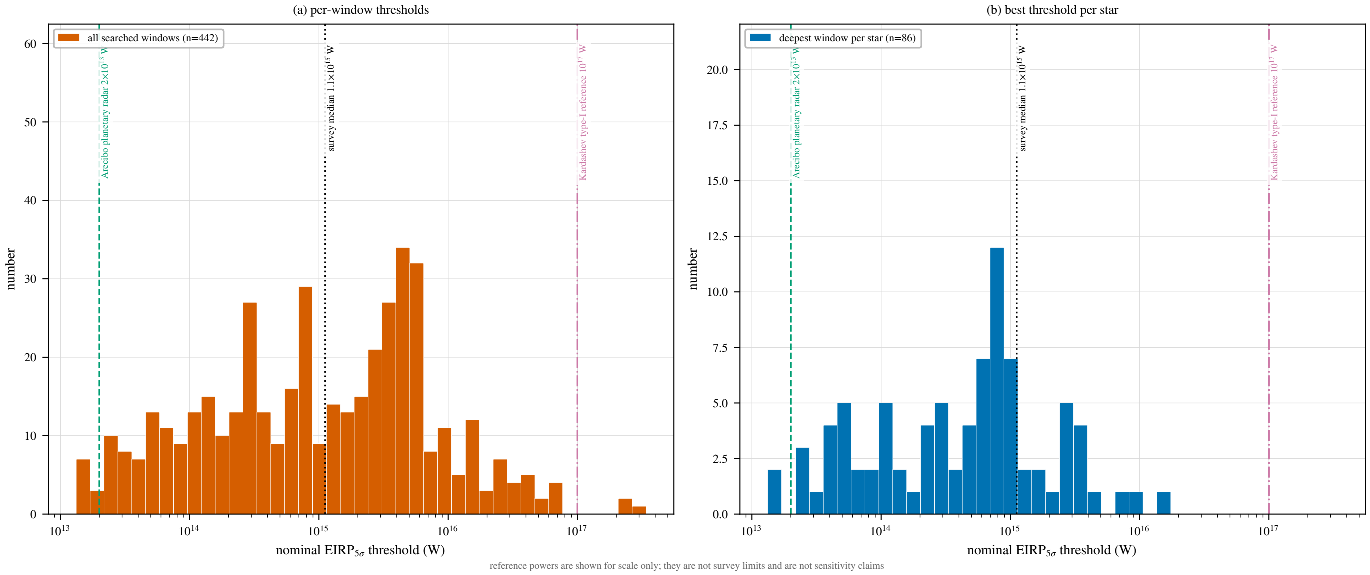

Fig. 1.— Literature context of this release's nominal thresholds. (a) Spectral power threshold ($\mathrm{W\,Hz^{-1}}$), the primary and like-for-like comparison: the same literature points carry over unchanged (one-Hz channels make their quoted EIRP and spectral power identical), while our kHz–MHz channels separate the two by up to six orders of magnitude – this survey (blue; nominal 5σ thresholds under the boxed reading rule of §4) against Enriquez et al. (2017) (orange diamonds), Margot et al. (2023) (aqua squares), Mason et al. (2024) (violet triangles). (b) Total transmitter power required for detection at each survey's native spectral resolution – not narrowband spectral sensitivity – with the deepest per-target limits of Price et al. (2020) as left-edge ticks; the grey dotted line marks EIRP $\propto d^2$, the green dashed line an Arecibo-like planetary radar (2×10^{13} W). Conventions differ among surveys (thresholds, drift coverage, confidence); context, not a ranking.

Six decades of searches divide into wide-field commensal surveys and targeted programmes on preselected stars, the modern generation dominated by Breakthrough Listen, whose GBT, Parkes, and MeerKAT programmes have published multi-thousand-star limits from 1.1 to 8 GHz (Isaacson et al. 2017; Enriquez et al. 2017; Price et al. 2020; Czech et al. 2026), with parallel efforts on FAST, LOFAR, NenuFAR, the Sardinia Radio Telescope, and commensal VLA, MeerKAT, and MWA back-ends (Zhang et al. 2020; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024); machine-learning rejection of RFI-dominated hit lists has become central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026) – a capability the classical threshold search we adopt lacks (§6). Two structural points shape the present survey. First, Margot et al. (2023) showed that a non-detection bounds the transmitter fraction only if the sample is representative of the underlying population – a condition interest-selected samples violate. Second, essentially all of this activity lies below $\sim$10 GHz: above it, the millimetre and submillimetre regime has been searched only once before, also with ALMA (Mason et al. 2024); sixteen programmes are tabulated in Appendix D.

Mason et al. (2024) searched archival Band 3 windows toward 28 "stellar bycatch" stars and placed $\mathrm{EIRP_{min}} > 7\times10^{17}$ W for the nearest, identifying the challenges of high-frequency work: drift rates scale with observing frequency, smearing erodes sensitivity unless a drift search is run, and the molecular line forest confuses. Our earlier pilot (White 2026; under review at the time of submission – the citation is provisional and will be updated on acceptance) extended that methodology to TRAPPIST-1 across Bands 3, 6, and 7 ($\mathrm{EIRP_{min}} \sim 5$–$10\times10^{13}$ W, no detection) and built the ranked 100-star list from which this survey grew. The incremental content of this submission is the census selection (§3), the 512-position control-ring false-alarm calibration, closure-phase vetting, the spectral-type-normalised continuum screen, the chirp and periodicity extension, and the cross-target frequency-occupancy check. No execution block is analysed in both papers: the pilot's 100-star search products are superseded, not re-used, here, and every window analysed in this paper was re-reduced and re-searched from the raw archive data with the pipeline of §4.

Figure 1 places this release's nominal EIRP thresholds among the published searches under two conventions. A quoted EIRP threshold is defined at the producing survey's channel width, and those widths span nearly six orders of magnitude between Hz-scale backends and ALMA's native correlator channels: panel (a) divides each limit by its native channel width to give $\mathrm{W\,Hz^{-1}}$, the like-for-like spectral comparison (the primary panel), and panel (b) is the total-power comparison – our limits are numerically comparable to the deepest L-band limits in panel (b) only because the nearest targets lie an order of magnitude closer, despite channels some 10^6 times coarser. Our linewidth convention is set out in §4.

A non-detection must be translated into a quantitative exclusion statement. Wright et al. (2018) formalised the "Cosmic Haystack" search volume; Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction; and Sheikh et al. (2025a) asked instead where Earth's own technosignatures would be detectable – a calibration point any survey's sensitivity can be read against.

The case for a census over interest-selected lists follows directly: most surveys select by proximity, spectral type, planets, or Earth Transit Zone membership, which is reasonable per unit time. The ALMA-specific form is disc selection: ALMA-detected discs are orders of magnitude brighter than the Solar System's zodiacal cloud would appear at interstellar distances, so disc-selected samples skew young and dynamically active. This survey therefore identifies every star within 40 pc in the usable field of at least one public ALMA observation (§3) – a bycatch census of the ALMA-observed subset of the local stellar population within a fixed volume, which is not the same thing as a census of that population. It removes technosignature-specific selection, not the astrophysical selection embodied in the archive itself; our 168 stars are ∼7 per cent of the known stellar population within 40 pc, and that residual coverage selection is characterised as a completeness function (§3, Table 3) rather than left implicit.

The coming decade will widen the search: the SKA (Siemion et al. 2015; Tremblay et al. 2026) and ngVLA (Ng et al. 2022) bring wide bandwidth and commensal architectures below ∼50 GHz, but neither offers ALMA's combination of resolution and collecting area above it in the near term, so a systematic mm/submm census remains an ALMA-specific measurement for some years.

3. SAMPLE SELECTION

Our sample comprises every star within 40 pc of the Sun (Gaia DR3 parallaxes, all distances direct inversions $d = 1/\pi$; every entry has $\pi \geq 25$ mas with $\sigma_\pi/\pi \leq 0.8$ per cent) with at least one public ALMA observation whose proper-motion-corrected position at the epoch of observation falls within the field of view of at least one execution block – not necessarily at the pointing centre, since mosaics and wide pointings serendipitously cover more than one catalogued star. No disc, planet, or spectral-type weighting is applied; those properties are descriptive metadata, not selection criteria. This identifies 168 unique target stars, processed nearest-first – a sensitivity-ranked order, not an unbiased draw, on which the prevalence statements of §6 are conditional.

The composition follows from this archival selection: just under two thirds of the 168 stars are M dwarfs (76 stars, 63 per cent), followed by 12 K, 11 G, and 8 F dwarfs, a single A-type star (β Pic), and six white dwarfs; six faint companions lack tabulated types. Fifteen of the 46 processed stars are confirmed exoplanet hosts (§6), and debris-disc hosts (β Pic, AU Mic, ϵ Eri, τ Cet) contribute continua that include circumstellar dust, which the continuum lane's spectral-type-normalised screen (§4) keeps from masquerading as an anomaly.

What the sample covers should be explicit: the 168 stars are every star within 40 pc with public ALMA coverage, so the census is complete with respect to the intersection of the 40-parsec reference catalogue with the public-archive snapshot (2026 September 8) that defines this release, and conditional on that intersection only. Against the reference Gaia DR3 catalogue of the solar neighbourhood (Figure 12, online-only appendix; 17 566 stars within 40 pc passing the parallax-quality cut of Table 15), the 168 stars are ∼1%. That reference list is itself incomplete at both ends, so the denominator is a convention, and we use the fraction only to say it is small. How the covered stars differ from the census is quantified in Table 15 (online-only appendix), so the selection function is inspectable rather than asserted: a strong enrichment of the nearest systems falling towards parity with distance, mild M-dwarf depletion with F/G enrichment, and planet-hosting in 20 of the 168 census entries. Age and disc-hosting flags are not uniformly available for the census and are documented qualitatively, with a machine-readable selection-function table part of the release.

ALMA coverage is itself strongly astrophysically selected – it inherits the proposers' interests – so any prevalence statement this survey makes concerns the ALMA-observed subset of the solar neighbourhood, not the neighbourhood as a whole. The stellar counts quoted throughout are catalogue-entry counts: the 120 entries comprise 113 distinct physical systems (seven designation-linked component pairs),[4] and because the 44 narrowband-covered entries include the spatially unresolved UV Ceti and AT Mic pairs, the 104 star/band datasets, 417 windows, and 57 execution blocks derive from 42 independent systems – the paired rows are bookkeeping, not independent trials.

"Usable public ALMA coverage" is frozen as a five-criterion decision tree, enforced for this release and not changed subsequently without an erratum: (i) at least one public execution block (EB) with QA2-calibrated visibilities covering the star's epoch-propagated position in the primary beam, with no minimum integration, sensitivity, antenna count, mosaic-tile count, or array configuration imposed beyond QA2 state itself; (ii) propagated position within one primary-beam FWHM of the field centre (largest accepted offset ∼0.7 FWHMs, Sirius B); (iii) each spectral window admitted individually, on the pipeline's quality vetting alone; (iv) no programme-category restriction – science, calibration, and observatory projects enter on identical terms; and (v) a fixed archive snapshot date (2026 September 8), re-evaluated only at release boundaries. No star or window was excluded by any criterion not listed here, and none for short integration (the shortest retained are 1.0 and 1.5 min).

Pointed versus serendipitous coverage is quantified from ALMA project metadata: almost every narrowband-covered star is the pointed science target in at least one of its execution blocks, the exceptions being a small number of blocks that point at another programme's science field or are calibrator-only executions – leaving Wolf 219 the sole genuinely serendipitous star. The residual selection function of an archival census is the union of individual ALMA target lists – a contrast with interest-selected technosignature samples that is one of degree, but of *quantified* degree: a bycatch fraction of 1 star in 44, 1 EB in 57.

4. DATA AND METHODOLOGY

Survey-specific vocabulary – target/band dataset, spectral window, execution block, control ring, candidate, molecular-line exclusion mask – is defined once in Table 11, Appendix B.

The design is summarised in Figure 2: one shared archive-ingestion stage feeds two independent search lanes. For each target we mine the public ALMA science archive for all calibrated (or locally recalibrated) execution blocks

[4] Component designations follow SIMBAD; in the per-target file (Appendix I) each entry carries its own row and distance, and spatially unresolved pairs (e.g. the UV Ceti entries at 2.7 pc, whose fluxes are blended) are marked in the notes column.

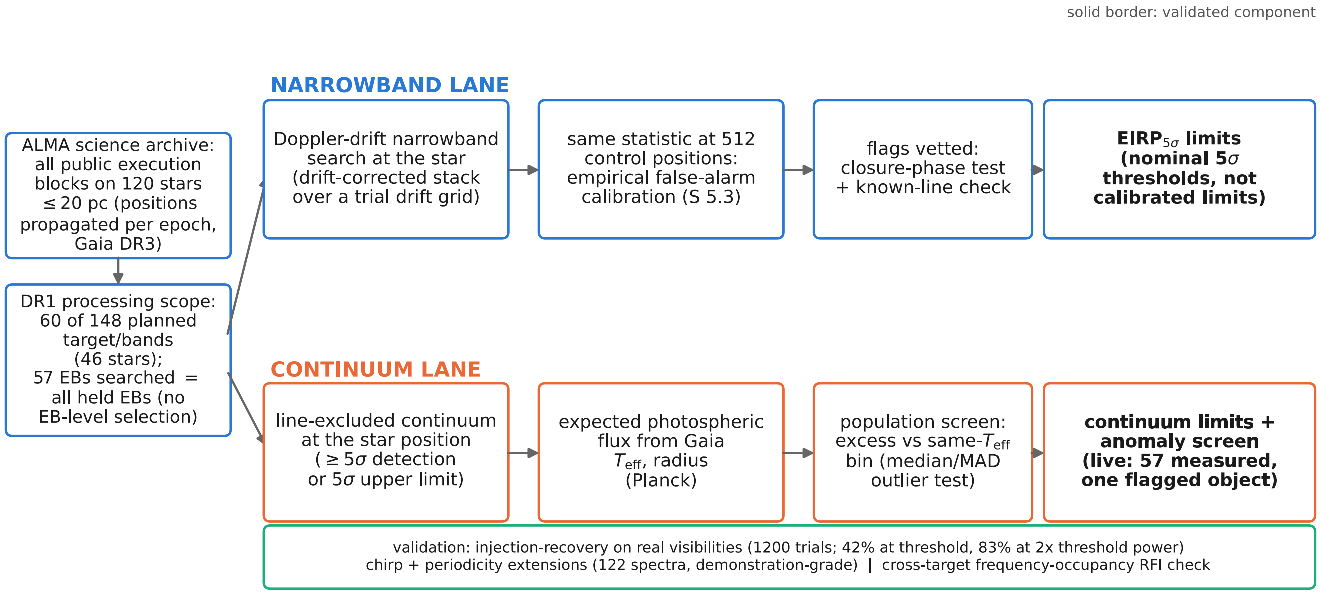

Fig. 2.— Design of the survey. All public execution blocks on the 168 targets are mined from the ALMA science archive, with positions propagated to each epoch from Gaia DR3, and searched along the two independent lanes described in the text. Solid borders mark validated components. The validation rail summarises the injection-recovery (1200 trials, Appendix E), chirp/periodicity, and frequency-occupancy analyses.

covering the star's Gaia-propagated position, within a bounded per-target processing budget, and search the resulting visibilities in (i) the *primary* lane – a narrowband, Doppler-drift-corrected search for technosignature candidates, following Mason et al. (2024) and Enriquez et al. (2017), converting each non-detection into an equivalent isotropic radiated power limit $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\min}\Delta\nu$, where $S_{\min}$ is five times the per-channel rms of the searched window (primary-beam corrected), $\Delta\nu$ its native channel width, and d the target distance; and (ii) an *ancillary* line-excluded continuum lane (Appendix L). Throughout, "5σ" denotes this local instrumental threshold in the sense of Table 2, calibrated separately and empirically (§5.4). The chirp/periodicity extension and the molecular-line census are defined in Appendices G and N; every limit, disposition, false-alarm calibration and completeness statement derives from the narrowband lane.

Both lanes exclude catalogued molecular transitions, at tolerances appropriate to their statistics. The continuum lane masks the eight common species (CO, ^{13}CO, C^{18}O, HCN, HCO$^+$, CS, CN, SiO) at native resolution before integrating; the narrowband lane applies a $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame velocity exclusion mask (§§4.2, 5.4) to any threshold crossing, and the same mask defines the effective searched bandwidth (≈87 of the 92.4 GHz unique frequency union). Masked regions reduce the frequency space a candidate can occupy, not the per-channel noise, so no threshold in the per-target file is affected by the mask.

All flux densities and EIRP thresholds are Stokes I quantities, so the limits apply to total transmitted power independent of polarisation geometry. What is lost is diagnostic, not sensitivity – polarisation is not used as a discriminator (no Stokes $V/Q/U$ extraction or circularity test is performed), and a polarisation-sensitive test is a natural future addition (§6.2).

How to read every limit in this paper. *Every EIRP$_{5\sigma}$ value here is a nominal, local instrumental detection threshold: five times the per-channel rms of the searched window, at that window's native channel width, for a transmitter at the star's position matching the searched temporal and spectral morphology – time-variable on track timescales, since phase-steady carriers dwelling in one channel for most of the track are suppressed by the statistic's own continuum subtraction (Appendix E). It is not a calibrated completeness limit: the injection campaign measures a recovery of roughly one half at the threshold itself (EIRP$_{50}$ = 1.10 EIRP$_{5\sigma}$ for the searched class), rising to 83 per cent at twice the threshold power (EIRP$_{90}$ > 2 EIRP$_{5\sigma}$; Appendix E). It is a total-power threshold set by ALMA's coarse native channelisation, not a Hz-resolution narrowband sensitivity. The same reading applies to every subsequent "5σ" and to the per-target file (Appendix I); it is stated once here and not repeated at each occurrence.*

4.1. *The search statistic, and the words used for its output*

TABLE 1
The statistical data path, in order. Steps 1–8 are executed identically at the stellar position and at all 512 control positions; only step 9 onward distinguishes them.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-channel subtraction of the time median
5	stack along each trial linear drift rate
6	robust per-position noise scale $\sigma(\mathbf{x})$ (MAD)
7	$T(\mathbf{x})$ from Eq. 1
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_\star \geq 5$ (a hit)
10	rank test: $T_\star > \max_i T_i$ (spatially significant hit)
11	molecular-line mask, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar frame
12	vetting: recurrence, closure phase, cross-target occupancy

TABLE 2
Nomenclature for search output and for the two senses in which a probability is quoted.

Term	Meaning
hit	one channel×drift cell at the stellar position with $T \geq 5$
spatially significant hit	a window whose $T_\star$ exceeds all 512 control statistics
candidate	a spatially significant hit surviving every veto; none in this release
nominal threshold	5σ local instrumental threshold, *not* a completeness limit
completeness	measured recovery of injected signals by the unmodified pipeline
$p_{\mathrm{rank,min}}$	$1/(N_{\mathrm{ctrl}}+1) = 1/513$, the rank *resolution* of the control ensemble
P(false flag)	the survey false-positive probability after thresholding, correlation and non-Gaussianity; not the same quantity

The primary statistic is formed on extracted spectra, not on images or on visibilities directly. For a sky position $\mathbf{x}$, the per-channel time series is median-subtracted along the time axis, the residual dynamic spectrum is stacked along each trial linear drift rate $\dot{\nu}$, and the statistic is the largest resulting single-channel amplitude in units of that window's noise,

$$T(\mathbf{x}) = \max_{\dot{\nu},\,\nu} \frac{S(\mathbf{x}, \nu, \dot{\nu})}{\sigma(\mathbf{x})}, \tag{1}$$

where S is the drift-stacked flux density in one native channel and $\sigma(\mathbf{x})$ is a single robust scale per position and window, estimated as 1.4826× the median absolute deviation of that position's median-subtracted spectrum over all channels, so it is local in position and global in frequency. $T_\star \equiv T(\mathbf{x}_\star)$ at the proper-motion-propagated stellar position; the identical operation, with the identical drift grid, channelisation and noise estimator, is applied at each of the 512 control positions to give $\{T_i\}$. Table 1 sets out the whole data path in order, so that the statistic can be reimplemented without recourse to the appendices.

Three words are used for the output, in the strict nesting of Table 2: *hit*, *spatially significant hit*, *candidate*. No object of this release is a candidate.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default (12–$13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$), the literature fiducial of Mason et al. (2024) with a threefold safety margin – a design boundary, not a claim that faster drift is physically implausible – and (b) where the target hosts confirmed exoplanets, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter co-located with a short-period planet cannot drift out of the searched window. For scale, $\dot{\nu}$ at frequency ν corresponds to a line-of-sight acceleration $a_{\mathrm{los}} = c\dot{\nu}/\nu \approx 3.9\,\mathrm{m\,s^{-2}}$ ($\sim$0.04 au around a solar-type star); Earth-motion contributions ($\lesssim 0.13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ combined) are two orders of magnitude below it and subsumed by the grid. These are *apparent* topocentric drift rates, including any deliberate transmitted chirp as well as acceleration-induced drift, which the survey does not attempt to distinguish – the constraints of this paper concern linear apparent drift within the searched range only.

Across the 417 windows the maximum searched drift rates span 1.1–$5.9\,\mathrm{kHz\,s^{-1}}$, the native channel widths 15.3 kHz to 31.25 MHz (median 15.625 MHz; Appendix I), and the trial drift rates per window 2 to 1944 (median 4). The threefold margin is consequential: at 1× the release would have been blind to two of its eight threshold-crossing windows, including the AU Mic candidate; at 5× no disposition changes – so transmitters drifting faster than $\sim 13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ remain unconstrained. Intra-integration smearing is negligible almost everywhere: at most 1.10 channels, median 0.001; only three windows have $\eta_{\mathrm{smear}} < 0.99$ (marked * in Appendix I; Table 12, Appendix C), the worst being the β Pic Band 6 fine window (effective threshold 1.74× nominal). The channel widths are the effective resolutions (verified against the measurement sets' own spectral window tables; Appendix C).

4.2. *Frequency and velocity reference frames*

Every frequency axis in this paper – searched spectra, tabulated ranges, candidate frequencies – is *topocentric*: the frequencies delivered by the ALMA correlator, with no velocity-frame shift applied to any data product at any stage.

Velocities enter at one place only, the stellar-frame line-exclusion mask of the vetting stage, whose conversion chain is given in Appendix C. The stacking behind the search statistic is coherent in frequency and in the drift-rate trial but preserves no electric-field phase coherence between integrations.

The native channel width is also the linewidth convention under which every EIRP threshold must be read – the commonest source of confusion in comparing technosignature limits across surveys. The quoted $\mathrm{EIRP}_{5\sigma}$ applies to *any* transmitter narrower than the native channel; dividing it by the channel width gives the minimum detectable spectral power for wider ones. Channelisation is the dominant price the archive exacts, but we deliberately do not convert it into a projected Hz-resolution sensitivity. A radiometer-equation scaling $P_{\min} \propto \sqrt{\Delta\nu}$ applied across six orders of magnitude in spectral resolution is a thought experiment, not an achievable ALMA sensitivity, and quoting it invites exactly the misreading this paper is trying to avoid. Thresholds are reported at the native channel width throughout. The worked example, the sub-channel qualifications, the smearing table and the channel-width verification are in Appendix C.

The narrowband search covers every distinct spectral window configured for an observation, up to a cap of 8 windows per target to bound compute cost. The cap bound exactly one target/band (61 Vir Band 7: four of eight windows searched; Appendix I), so the frequency-space selection function it introduces is fully tabulated rather than hidden. Of the 104 star/band datasets, 53 have every configured window searched; the remaining 3 (τ Cet, Kapteyn's Star, Van Maanen's Star) are represented by their deepest single window, marked in Appendix I, with full multi-window coverage deferred to the next release. Each searched window yields its own $\mathrm{EIRP}_{5\sigma}$ limit as a separate row of the file (Appendix I); Figure 6 plots only the deepest result per target/band.

Velocity coincidence, not drift, is the operative discriminant when a threshold crossing is checked against the eight masked species (§4): circumstellar emission sits at rest offset by the gas's own kinematics, whereas a technosignature is free to appear at any frequency and drift rate. The searches as executed used a narrower one-channel-width implementation that agrees with the velocity-space form on every disposition here; the tolerance's justification and the bandwidth it removes are in §5.4.

Both the narrowband thresholds and the continuum fluxes are corrected for ALMA's primary-beam response at the star's offset from the phase centre. For most targets the correction is negligible (median factor 1.0001, 90th percentile 1.012, none above 2%). Two datasets sit at large offset. ϵ Eri Band 6 lies at $\sim$0.7 primary-beam FWHM, where the correction reaches 2.9–3.4$\times$ and the Gaussian beam form is not a valid approximation; rather than publish a provisional value we *withhold* those four windows from this release, and they enter no number in this paper. They are released, flagged, in the per-target file, and return once the holography beam model has been applied. Sirius B is retained: its corrections are 3.6–16 per cent across bands, a bounded systematic rather than an invalid model, and it is quoted as such. Excluding Sirius B as well would change neither endpoint of the quoted EIRP range nor the median ($1.3 \rightarrow 1.4 \times 10^{15}$ W) nor any conclusion.

4.3. *Reference transmitter benchmarks*

For physical context the EIRP thresholds of §5 are compared throughout to two human-transmitter benchmarks, both from Sheikh et al. (2025a): an Arecibo-like planetary radar (S-band EIRP 2×10^{13} W, the most powerful deliberate directed transmitter an Earth-equivalent civilisation could point at us, used purely as a technological-power benchmark), and unintentional Earth mobile-network leakage with no signalling intent (EIRP of order 4×10^{9} W).

4.4. *Ancillary and validation analyses*

Four analyses beyond the narrowband search are executed in this release, each specified in full online (Table 14) and summarised here only so far as the main text depends on it.

Injection-recovery validation (Appendix E). Synthetic narrowband tones of known amplitude were injected into real calibrated visibility data and recovered with the unmodified pipeline (1200 trials across six window configurations, three bands, and both resolution classes). Two outcomes carry into the limits. First, the per-channel time-median subtraction absorbs phase-steady carriers, totally so on the coarse (15.625 MHz) windows (0 of 500 coarse injections recovered from 4σ to 10σ), so the coarse-window limits bound transmitters time-variable on track timescales rather than phase-steady carriers. Second, for the drifting class on the fine window recovery is 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ, giving $\mathrm{EIRP}_{50} = 1.10\,\mathrm{EIRP}_{5\sigma}$ and, as a bound, $\mathrm{EIRP}_{90} > 2.0\,\mathrm{EIRP}_{5\sigma}$ (Fig. 11, Table 17). Every table here stays in nominal $\mathrm{EIRP}_{5\sigma}$; the occurrence limit of §6 is the one place the measured completeness is folded in numerically, and Band 8 completeness is transferred by shared construction rather than measured.

Drift-ceiling caveat. The searched ceiling is a fixed $12\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, i.e. a line-of-sight acceleration $|a| = 3.6\,\mathrm{m\,s^{-2}}$, uniform across every window. This clears Earth-rotation and Earth-orbit drifts by about two orders of magnitude, but it sits 11 per cent *below* the $4.0\,\mathrm{m\,s^{-2}}$ appropriate to a close-in planet on a TRAPPIST-1 b-like orbit. Carriers on the most rapidly accelerating bodies are therefore not fully covered, and the limits of §6 are conditional on that ceiling (Figure 13).

Closure-phase vetting (Appendix F). Threshold crossings face an automated point-source test that single-dish and beamformed surveys lack, since it requires multiple independent baselines; it is one-sided – evidence against celestial point-source behaviour, never an authentication. It is implemented and simulation-validated, but at this survey's flux densities it has *no discriminating power*: per-baseline S/N never exceeds 5.1 in the positive-control fields, so no flagged window of this release could have failed the test whatever its true nature. We therefore report it as a proof-of-principle diagnostic for future, deeper releases and not as an operative filter here, and we do not describe any window as having

TABLE 3
The complete state of the survey as frozen for this release. Every count quoted anywhere in this paper reconciles with this table; where a number refers to a subset, that is said at the point of use. Thresholds are nominal in the sense of the boxed rule of §4.

Quantity	Value
Census stars within 40 pc	168
Stars searched; independent systems	85; 79
Star/band datasets	104 of 208 planned (50%)
Spectral windows searched	417
extracted; repeats, defective, withheld	448; 21, 6, 4
fine (<5 MHz) / coarse channels	111 / 306
Execution blocks	100
Distance range	1.30–38.83 pc
Unique frequency union	92.4 GHz (34 intervals, 89.6–495.1 GHz)
Effective after line-exclusion mask	≈87 GHz
Native channel widths	15.3 kHz–31.25 MHz (median 15.625 MHz)
Drift-rate ceilings	1.1–5.9 $\mathrm{kHz\,s^{-1}}$ (12.0–13.3 $\mathrm{Hz\,s^{-1}\,GHz^{-1}}$)
Per-window on-source time	21–5274 s (median 2177 s)
Exposure×bandwidth	1.4×10^{6} s GHz
Nominal $S_{\rm min}$	0.48 mJy–1.30 Jy (median 6.7 mJy)
Nominal $\mathrm{EIRP}_{5\sigma}$	1.6×10^{13}–2.8×10^{17} W (median 1.3×10^{15})
Measured recovery at threshold (searched class)	42^{+16}_{-14}%
Hits ($\geq5\sigma$ at the stellar position)	18 windows
Spatially significant hits	4 windows
circumstellar or foreground CO	3
consistent with the control ensemble	1
Credible technosignatures	0

"survived" it. The flagged windows are formally consistent with a point source, which under these conditions carries no evidential weight.

Extended and cross-target screens (Appendices G and H). A chirped (quadratic-drift) extension and a per-channel periodicity search were run against the 122 retained spectra as a demonstration rather than a survey-grade screen; their null behaviour is as expected (star-exceeds-ensemble rates 10.7 and 13.1 per cent against an 11-per-cent chance rate), no instance is a candidate, and no limit derives from them. Separately, threshold crossings towards different targets were cross-matched for recurrence at the same observed sky frequency – the archival substitute for the ON/OFF cadence this single-pointing survey lacks. Of the 366 star pairs whose searched frequency ranges overlap, none falls within the matching kernel, subject to one bookkeeping caveat (64 crossings of one β Pic window untested) recorded in the appendix.

5. RESULTS

The survey is strictly volume-ordered (nearest star first; §3), so these results are weighted towards the very nearest stars, spanning 1.3–19.6 pc. Full per-target values, including each band's exact searched frequency range, are given in the online-only per-target file (Appendix I).

5.1. *Per-star summary*

Table 23 (online-only) gives the one-line-per-star view complementing the per-window per-target file (Appendix I). Two stars of the release are continuum-only. All values are read directly from the per-target file: EIRP values are nominal native-channel thresholds under the boxed rule of §4, and continuum fluxes are primary-beam corrected.

5.2. *Barnard's Star and Wolf 359*

Barnard's Star and Wolf 359 – the second- and fourth-nearest stellar systems – have not, to our knowledge, previously been searched for technosignatures at ALMA frequencies. Both yield non-detections from public Band 6 archival data: $\mathrm{EIRP}_{5\sigma}$ limits of 6.15–6.89×10^{13} W (Barnard's Star, 4 windows) and 7.84–9.10×10^{13} W (Wolf 359), with continuum upper limits of 0.575 and 0.453 mJy.

5.3. *Data-quality exclusions and open items*

Status of the 208 planned target/bands: 104 searched to completion and reported here, the remainder queued for later releases, and a small number attempted and failed at the process level and excluded from all counts. Four classes of entry within the 104 are absent from the released per-target file (Appendix I) and are itemised with their forensics in Appendix R: (i) four process-level search failures, each still carrying a continuum non-detection; (ii) one pair excluded for data quality; (iii) windows partially withheld for a noise-handling defect; and (iv) one relabelled entry.

Item (iii) sets this paper's headline depth. A reproducible defect produces implausibly small noise in a minority of coarse-channel windows. An initial implementation excluded the two affected windows by name. Extending the sample showed that a name-based exclusion does not generalise, so the defect is now caught by a physical test. For a correctly weighted measurement the product $\sigma\sqrt{t_{\rm on}\,\Delta\nu_{\rm ch}}$ is set by system temperature and collecting area, so it varies by factors

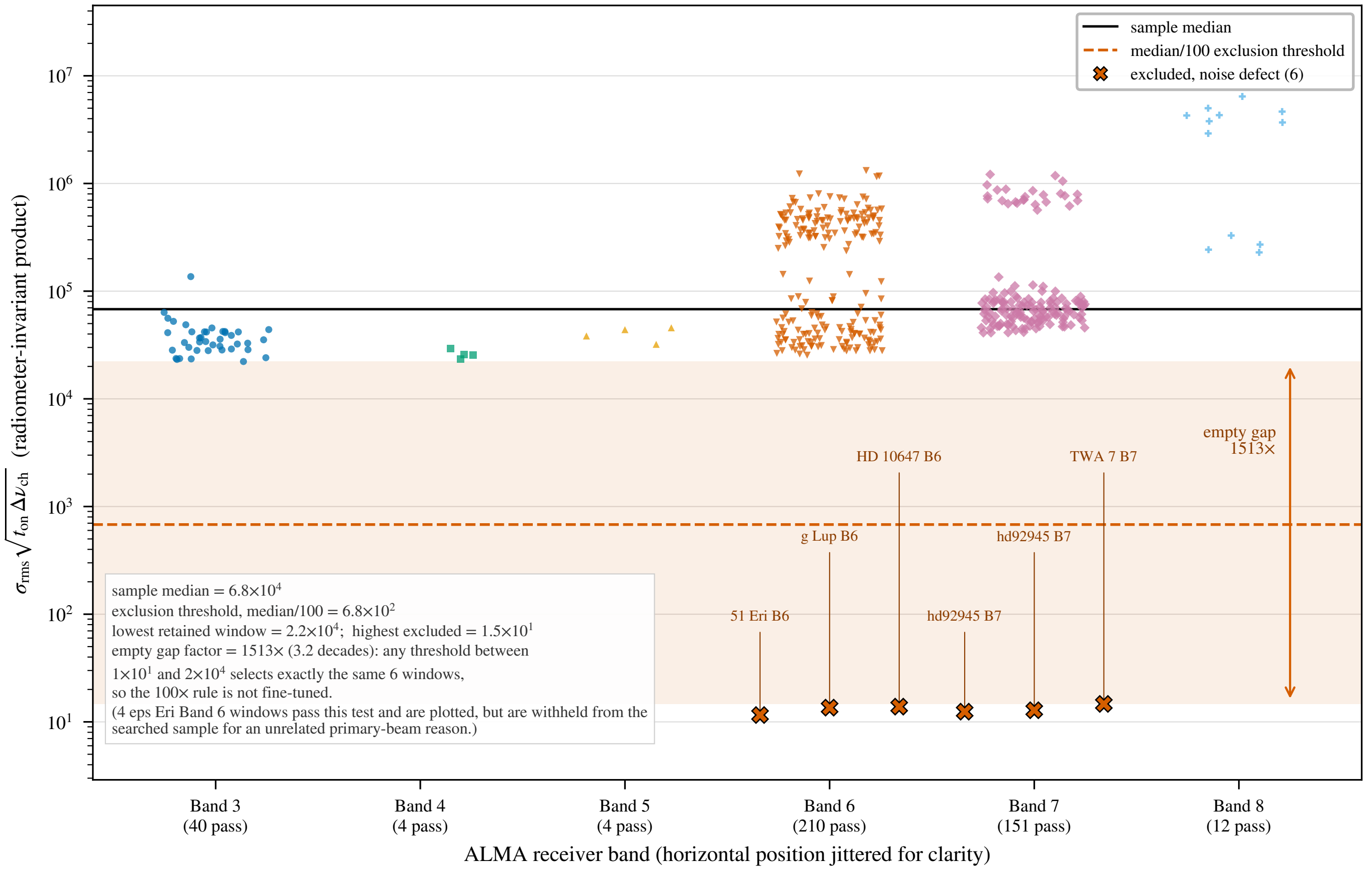

FIG. 3.— Quality control on the noise estimate (§5.3). The radiometer-invariant product $\sigma\sqrt{t_{\rm on}\Delta\nu_{\rm ch}}$ varies by factors of a few across a heterogeneous archive; the six excluded windows lie a factor 1.5×10^3 below the lowest retained window, separated by an empty gap.

of a few across a heterogeneous archive; windows falling more than 100× below the sample median are excluded. Six of 417 windows fail, across five stars, lying a factor 1.5×10^3 below the lowest retained window across an empty gap of 3.2 decades (Figure 3), so the threshold is not fine-tuned: any value between 1.5×10^1 and 2.2×10^4 selects exactly the same six windows. A second bookkeeping rule needs stating because it is not purely mechanical: 21 of the 448 extracted rows repeat a (star, execution block, frequency range, channel width) key, and we keep one row per key. Most such pairs are identical measurements, but two are not – two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, and the two HR 1010 Band 6 rows straddle the 5σ threshold (5.10 and 4.95) – so the count of windows containing a hit is rule-dependent at the level of one window. The four spatially significant windows are not: no discarded row is both a hit and star-beats-ring. Every extracted row, kept or discarded, is released with its status. Their nominal thresholds, 2.0–7.8 × 10^{12} W, would otherwise have set this paper's headline depth; excluding them, the best threshold is 1.6×10^{13} W. The supporting audit is otherwise clean apart from four windows of elevated system temperature, and every census identifier re-resolves independently through SIMBAD and *Gaia* (Appendix R).

5.4. *Technosignature search*

All 104 star/band datasets of the searched sample yielded at least one narrowband result; the datasets that failed at the process level are itemised in §5.3 and are not counted among them. They span 417 spectral-window measurements after removal of 21 duplicate rows found in the catalogue audit (Appendix I). The $\mathrm{EIRP}_{5\sigma}$ limits span 1.6×10^{13}–2.8×10^{17} W, median 1.3×10^{15} W (Figures 6, 7 and 1). As expected for a volume-ordered survey, the deepest limits are for the nearest systems (1.6×10^{13} W for the UV Ceti pair at 2.7 pc); the shallowest is η Corvi, the first Band 8 target searched, where distance, coarse channels, and receiver performance set the survey's current ceiling. Against the benchmarks of §4.3, only targets within ∼3 pc reach below the Arecibo-like planetary-radar EIRP, and none approaches the $\sim 4 \times 10^9$ W of unintentional Earth leakage.

Four windows contain a *spatially significant hit*. None survives vetting, so this release reports *zero candidates*: Table 4 follows the decision cascade and Table 5 gives each flagged window and the evidence that dispositioned it.

Two questions are kept separate: whether four spatially significant hits is more than this survey's own calibration predicts (a counting question), and whether any individual hit is distinguishable from the empirical background or from known astrophysical emission (the dispositions below). The answers are no and no.

SURVEY-LEVEL FALSE-ALARM CALIBRATION

TABLE 4
The narrowband decision cascade of this release. Counts are for the 417 searched windows after quality control (§5.3); the vocabulary is that of Table 2.

Stage	Count
Channel × drift trial cells (all windows)	3.97×10^7
Control positions compared per window	512
Windows searched after quality control	417
Windows containing ≥1 hit ($\geq 5\sigma$ at the stellar position)	18
Windows containing a spatially significant hit	4
attributed to circumstellar or foreground CO	3
indistinguishable from the control ensemble	1
Candidates (surviving every veto)	0

TABLE 5
The four windows containing a spatially significant hit, and the evidence that dispositioned each. $T_\star$ is the search statistic at the stellar position (Eq. 1) and 'ring max' the largest of the 512 control statistics of the same window; p is the add-one rank probability, whose floor is $1/513 = 0.0019$. The velocity offset is topocentric, from the nearest catalogued transition of the eight masked species; the $\pm 50\,\mathrm{km\,s^{-1}}$ mask is applied in the stellar frame after the conversion chain of Appendix C. No disposition rests on a single test.

Window	ν (GHz)	$T_\star$	ring max	Δv ($\mathrm{km\,s^{-1}}$)	Disposition
β Pic B3	115.2605	14.65	7.27	−26.9 (CO 1→0)	circumstellar CO
β Pic B6	230.5164	11.68	9.12	−28.4 (CO 2→1)	circumstellar CO
HD 48370 B6	230.7305	27.10	26.83	−34.2 (CO 2→1)	CO cloud emission
CP−72 2713 B7	345.1152	5.81	5.68	−1326 (CO 3→2)	control-ensemble background

Each window-level test compares the peak signal-to-noise ratio at the star's position with the same statistic at 512 control positions. Because the drift-rate grid and the thousands of channels are applied identically to star and controls, this calibration absorbs the per-window trials factor empirically. From this version the statistic is *symmetric*: the on-source quantity is the single propagated stellar position, exactly what each control contributes. Earlier versions maximised it over a small region centred on that position while each control remained a single position, and that asymmetry was not benign. Under a pure-noise null with exchangeable positions the maximum of $n_{\rm src}$ source-region positions exceeds that of $n_{\rm ctrl}$ single controls with probability $n_{\rm src}/(n_{\rm src} + n_{\rm ctrl})$, which for the $n_{\rm src} = 135$ region and 512 controls of a representative window is 21 per cent; the measured region-max star-beats-ring rate, 64 of 417 windows (15.3 per cent; Wilson 95 per cent interval 12.2–19.1), is of that order. The symmetric statistic instead gives star-beats-ring in 5 of 417 windows (1.2 per cent; Wilson interval 0.5–2.8), consistent with the exchangeable expectation $1/513 = 0.19$ per cent per window inflated by the real maps' heavy tails, and stable against the rate measured on the ≤20 pc subsample alone. The exchangeability rate is thus the operative one, and the survey-level false-alarm expectation follows from it directly rather than through the bookkeeping ladder the asymmetry previously required.

A candidate is gated by the joint criterion, whose threshold leg dominates the rate: only 18 of 417 windows contain any $\geq 5\sigma$ crossing at the stellar position. Each window contributes at most the exchangeable rate $1/(1+n_{\rm ctrl}) = 1/513$, so the pure-noise null predicts $417/513 = 0.81$ flags. Four flagged windows is more than that budget alone would produce ($P(\geq 4 \,|\, 0.81) = 0.9$ per cent), which is the correct reading rather than a tension: three of the four are identified astrophysical emission and were never chance events. What the chance budget must explain is the residual – one unattributed crossing against 0.78 expected across the 402 non-β Pictoris windows, the order of the expectation. This is the operative survey-level false-alarm estimate; it absorbs execution-block systematics into the fixed block margins and assumes only that, within a block, crossing status is exchangeable with star-beats-ring status. Secondary bookkeeping models span 1.0–9.2 per cent (Appendix P); the dominant residual uncertainty is the star-beats-ring rate itself (Wilson 95 per cent interval 12.2–19.1 per cent), giving a 4–19 per cent bracket on $P(F \geq 3)$.

One ensemble limit follows. With 512 controls the smallest resolvable per-window p-value is 1/513, so a Bonferroni threshold over 417 windows (1.1×10^{-4}) lies below what the controls can measure. They can show a crossing to be unremarkable but cannot certify one as significant survey-wide; that burden falls on localisation, recurrence, and astrophysical attribution, which decide each flag below. We state the consequence for the planned sequence of releases now rather than after the fact. Each release of N windows contributes $\sim N/513$ chance flags, so a programme reaching the full ∼2000-window census expects ∼4: reporting one unattributed marginal crossing per release is the *expected* outcome, not an accumulating anomaly. We therefore pre-commit to the criterion that no crossing will be advanced as a candidate on rank statistics alone at any point in this programme. Promotion requires evidence the control ensemble cannot supply – recurrence at an independent epoch, or a control ensemble enlarged for that window until the rank probability can resolve a survey-wide significance ($\gtrsim 10^4$ positions for a Bonferroni threshold over the completed census; §6.2). Until then every such crossing is reported, dispositioned as consistent with chance, and released with its products. No model in the bracket promotes any flagged window to a candidate: the three CO dispositions rest on kinematic attribution to emission that is resolved, mapped, or independently published, and the CP−72 2713 standing is a rank statement invariant to the bookkeeping model.

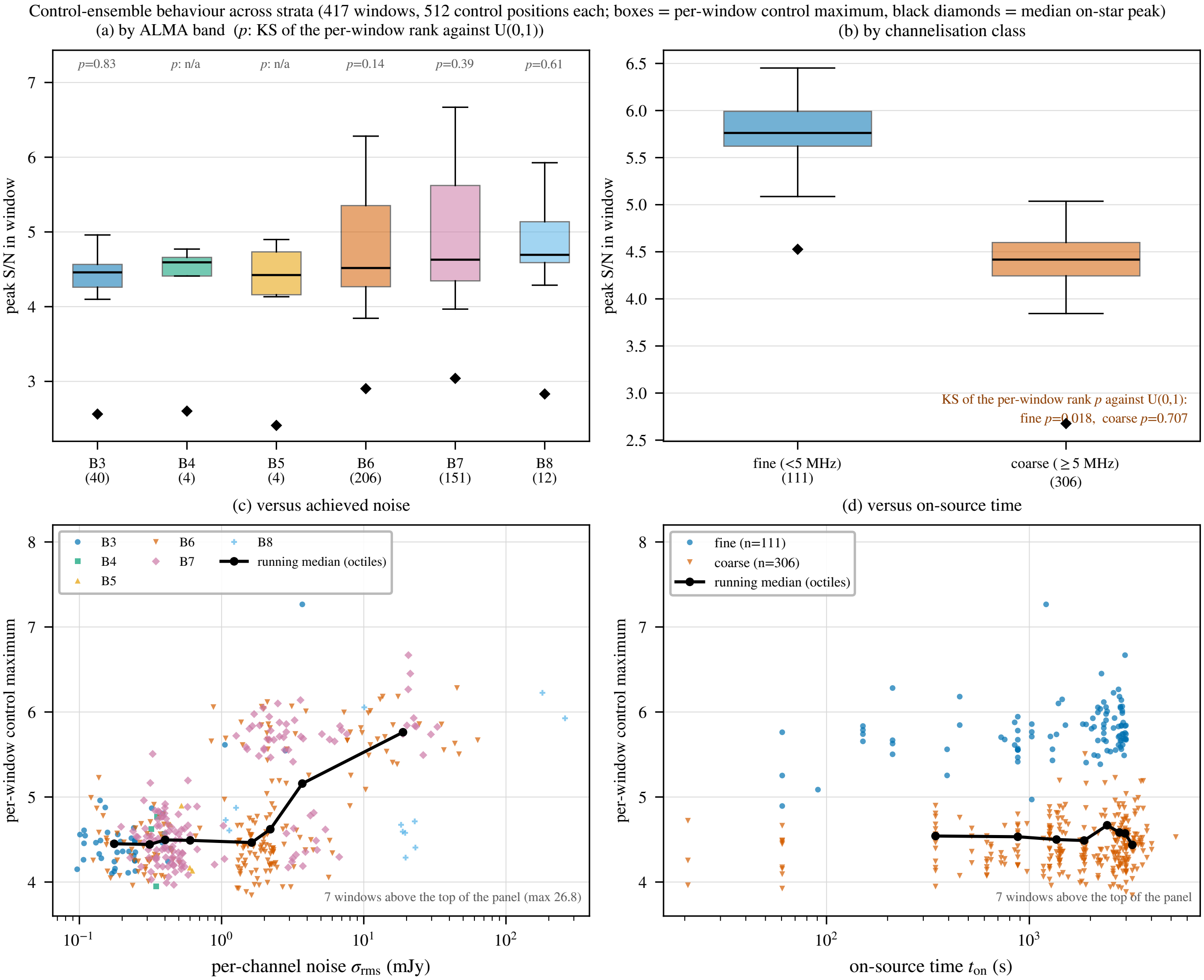

Fig. 4.— Exchangeability of the 512-position control ring, by stratum. Boxes give the per-window control maximum, diamonds the median on-star peak, and p the Kolmogorov–Smirnov test described in the text. The ensemble tracks the on-star statistic in every band and at both channelisations; the fine-channel stratum is the one departure ($p = 0.018$), driven by the four flagged windows of Table 5.

Is the ring exchangeable with the star? The 1/513 expectation holds only if the star and its 512 controls are exchangeable under the null, which spatially varying noise, sidelobes, correlated pixels or phase-centre-localised calibration residuals could each spoil. We measure it by stratum (Fig. 4). The median per-window control maximum is flat across receiver bands – 4.46, 4.59, 4.42, 4.52, 4.63 and 4.69 in Bands 3 to 8 – and a Kolmogorov–Smirnov test of the per-window rank of the star among its own controls against $U(0,1)$ is consistent in every band ($p = 0.14$–0.83) and for the coarse-channel stratum ($p = 0.71$). Fine-channel windows sit systematically higher in the statistic (control maximum 5.76 against 4.42), which is expected from their far larger channel×drift trial count and is not itself a violation, because the test is conditional on the window.

One stratum does depart, and we report it rather than pool it away: the fine-channel windows fail the same $U(0,1)$ test at $p = 0.018$ ($n = 111$), and contain four windows whose on-star peak exceeds their own control maximum against 0.22 expected. Those four are precisely the four flagged windows of Table 5. The departure is therefore astrophysical signal concentrated in the windows with the finest channels, where real narrow lines are detectable, rather than evidence that the control ring misbehaves; but it is a genuine stratum-level departure from exchangeability, and a reader should treat fine-window rank probabilities as slightly optimistic. Two stratifications cannot be made from the retained products, which store no per-control radial position or primary-beam gain: the ring sits at a radius small compared with the primary beam (targets lie a median 0.1 arcsec from the pointing centre), so all controls share the target's beam response by construction, but we have not measured that and say so. The remaining robustness checks – the trials-count reconciliation under channel adjacency and the control-ring geometry – are in Appendix P.

THE SYMMETRIC STAR-VERSUS-CONTROL NULL

The survey's designed principal test applies the identical source-region, drift-rate, and frequency maximisation at the star's position and at every one of the 512 control locations under a single search rule. That requirement cannot be met from the frozen release products, so its domain is the seven locally reprocessed windows – the six validation windows and the AU Mic candidate's first-epoch block – all of which give $T_\star = 3.7$–4.9 against control maxima of 6.7–7.7, $p = 0.81$–1.00 (Figure 10, online-only appendix): the star position is statistically exchangeable with its control ring wherever the test can be run at full symmetry. Extending it to all 417 windows requires per-window raw-visibility reprocessing that the retained products do not support, so the empirical star-versus-ring calibration above remains the operative release-wide estimate and the symmetric statistic a validation subset. The machinery ships with the release; the full statement, with the consolidated family-wise trials factors of Table 16, is in Appendix P.

DISPOSITIONS OF THE FOUR FLAGGED WINDOWS

Two of the four are towards β Pictoris, a debris-disc system with previously reported CO emission (Matrà et al. 2017): one in Band 3 at 115.2606 GHz, 10.56 MHz below the CO(J=1→0) rest frequency, the other in Band 6 at 230.5161 GHz, 21.87 MHz below CO(J=2→1). Neither coincidence is exact, so the automated line check did not catch them. Carried through the full frame chain (Table 6; Fig. 5), the offsets correspond to the same kinematic component – -2.7 and $-3.5\,\mathrm{km\,s^{-1}}$ from stellar systemic, $0.74\,\mathrm{km\,s^{-1}}$ apart – in the two lowest CO rotational transitions of a system with directly detected circumstellar CO; the coarse Band 6 window without a candidate, from a third epoch six years earlier, independently shows CO at $-3.6\,\mathrm{km\,s^{-1}}$ in the same frame. All three epochs and both transitions land within $0.9\,\mathrm{km\,s^{-1}}$ of each other and $3.6\,\mathrm{km\,s^{-1}}$ of systemic, deep inside the $\pm 50\,\mathrm{km\,s^{-1}}$ mask. The frame chain matters: the raw topocentric offsets are -27.5 and $-28.4\,\mathrm{km\,s^{-1}}$. We attribute both to this kinematically offset CO emission.

Their empirical significance under the release background is nonetheless extreme, so the attribution should not be mistaken for dismissal: the Band 3 region peak of 20.3 exceeds all 417 per-window control maxima (add-one $p < 0.0044$), and the Band 6 peak of 12.8 exceeds all but one – that window's own CO-filled control ring, which reaches 15.2, by far the highest control maximum of the release ($p \approx 0.0088$). Standing out this sharply is why they demanded astrophysical attribution rather than summary rejection as noise. That attribution came from inspecting the offsets against known CO kinematics, not from the automated chain, which did not catch them; the alternative would require a transmitter placed coincidentally at matched offsets from two CO transitions in the same star.

The third, towards HD 48370 in Band 6, is the largest excess in the release in absolute terms and the smallest in relative terms: $T_\star = 27.10$ against a control-ring maximum of 26.83, so the star exceeds its ring by one per cent of the statistic. That signature – star and controls rising together – is what emission filling the primary beam looks like, and the window sits $-34.2\,\mathrm{km\,s^{-1}}$ topocentric from CO(J=2→1). The attribution is independently published: Cataldi et al. (2023) report prominent CO(2→1) and ^{13}CO(2→1) emission towards HD 48370 at $v_{\mathrm{bary}} \approx 41\,\mathrm{km\,s^{-1}}$, well separated from that system's $23.6\,\mathrm{km\,s^{-1}}$ systemic velocity and attributed by them to foreground cloud contamination; our window recovers that feature. The case is also a stated limitation of the spatial test: what identifies this window is the velocity coincidence and the near-equality of star and ring, not the rank statistic.

The fourth, towards CP$-$72 2713 in Band 7, is the only flagged window with no astrophysical attribution, and it is marginal. $T_\star = 5.81$ against a ring maximum of 5.68: it beats its ring by 0.13 and therefore attains only the rank-probability floor, $p = 1/513$, which is the smallest value 512 controls can express and not a measurement of significance. The margin is smaller than the scatter of the same statistic across this star's own windows, whose other fine-channel control maxima are 5.16, 5.41, 5.67 and 5.85: 5.81 is a value its control ensembles reach routinely in windows where the star does not cross threshold. The nearest transition of the eight masked species lies $1326\,\mathrm{km\,s^{-1}}$ away; since that mask is a contamination cut rather than a claim of completeness over the mm/submm line forest, we re-queried the window against a wider catalogue, whose closest entry is H^{13}CN(J=4→3) at $-195\,\mathrm{km\,s^{-1}}$, still far outside any circumstellar tolerance. No line attribution is available. The archive holds a single execution block covering this frequency, so no second-epoch recurrence test is possible. We report it as a marginal crossing consistent with the 0.81 flags expected by chance across the release, attach no technosignature interpretation to it, and note that a recurrence observation is the one measurement that would settle it.

LINE EXCLUSION: DEFINITION, COST, AND LIMITS

The searches ran with an automated line check matching a crossing against catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris candidates showed that check inadequate: both fell outside its one-channel tolerance (by 10.6 and 21.9 MHz) while corresponding, after the frame chain, to a kinematic component within $3.6\,\mathrm{km\,s^{-1}}$ of stellar systemic of a resolved, previously mapped disc – exactly the regime a channel-width tolerance is blind to. Before publication every crossing was reclassified with a velocity-space mask ($\pm 50\,\mathrm{km\,s^{-1}}$ in the stellar frame around each catalogued transition), which is the published pipeline's line-exclusion definition from this release onward. Its kinematic justification, its empirical conservatism, the retrospective reclassification of every crossing, and the "Schelling point" counter-argument that a deliberate communicator might choose exactly these frequencies are set out in Appendix J; applied retrospectively the mask rejects exactly the two β Pictoris candidates and changes no other disposition.

The masking cost is stated on one definition throughout: every JPL catalogued transition of the eight species (276 transitions), displaced by each system's radial velocity, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame half-width, intersected with each window and merged. The release effectively searches 87.0 GHz of unique sky frequency – the disjoint union of the

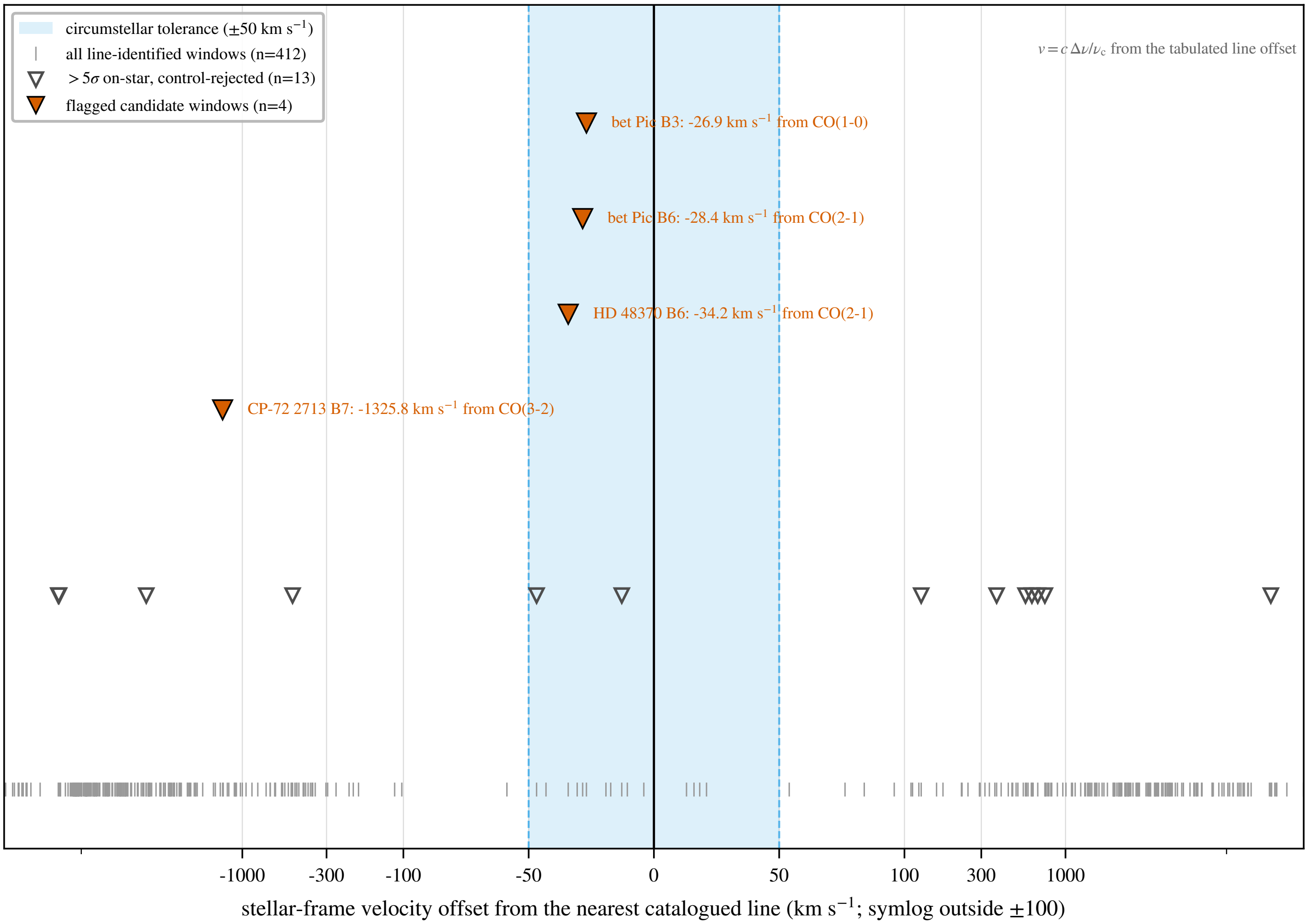

FIG. 5.— Stellar-frame velocity offsets of the candidate windows, carried through the full frame chain (topocentric → barycentric → stellar; Table 6). Shaded band: the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line exclusion mask. *Top:* the two β Pictoris candidates fall well inside the mask as a single kinematic component, resolved in the inset. *Bottom:* the flagged window towards CP−72 2713, and the AU Mic crossing that the symmetric statistic does not flag (§5.5), sit far outside any circumstellar tolerance and are dispositioned by their spatial behaviour, not by the mask.

TABLE 6
LINE-ATTRIBUTION AUDIT OF THE THREE CANDIDATE WINDOWS (FIRST THREE ROWS) AND THE COARSE β PICTORIS BAND 6 WINDOW WITHOUT A CANDIDATE THAT INDEPENDENTLY SHOWS THE SAME CO COMPONENT, CARRIED THROUGH THE FRAME CHAIN OF §4.2. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED REST FREQUENCY (CO J=1→0 115.271202 GHz, CO J=2→1 230.538000 GHz, PICKETT ET AL. 1998). THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME (BLOCK UIDS IN TABLE 10); THE SYSTEMIC RADIAL VELOCITIES ARE THE CATALOGUED VALUES (SIMBAD; *Gaia* DR3 FOR β PICTORIS, GAIA COLLABORATION ET AL. 2022; *Gaia* DR2 FOR AU MIC, FOUQUÉ ET AL. 2018).

Flag	UTC start	$f_{\rm obs}$ (GHz)	Transition	Topo. offset (MHz; $\mathrm{km\,s^{-1}}$)	Bary. corr. ($\mathrm{km\,s^{-1}}$)	$v_{\rm sys}$ ($\mathrm{km\,s^{-1}}$)	Stellar-frame offset ($\mathrm{km\,s^{-1}}$)
β Pic B3	2022-03-02 23:14	115.260644	CO(1→0)	−10.56; −27.46	−7.90	+16.84	−2.72
β Pic B6	2019-03-12 00:14	230.516128	CO(2→1)	−21.87; −28.44	−8.15	+16.84	−3.45
AU Mic B6	2014-08-18 04:01	230.825230	CO(2→1)	+287.23; +373.51	−10.02	−4.71	+378.82
β Pic B6†	2013-10-06 06:23	230.528134	CO(2→1)	−9.87; −12.83	+7.63	+16.84	−3.62

†Coarse Band 6 window without a candidate (Appendix I); shown because it independently exhibits the same CO component as the candidate fine window of the same band.

searched windows minus the part of the mask inside it (Figure 7) – down from 92.4 GHz before masking, a 5.7 per cent reduction. Summed over the 417 windows the mask removes 41.4 GHz of the 700.8 GHz of gross window bandwidth (5.9 per cent; per-species, per-band breakdown in Table 22, online-only appendix), and the per-window usable fraction has median 0.960 (10th percentile 0.790). Union bandwidth is exposure-blind, a window visited for seconds counting the same as one integrated for an hour, so we also report the summed per-star frequency union, 371.9 GHz, and the summed product of on-source time and searched bandwidth, 1.4×10^6 s GHz (per-window on-source time spans 12.1–5274 s, median 2570 s). Because the mask governs dispositions and not the noise-derived thresholds, the quoted $\mathrm{EIRP}_{5\sigma}$ values are unchanged; no signal was found outside the excluded velocity regions in any window.

TABLE 7
The AU Mic candidate frequency (230.8252 GHz) under the three treatments that decide it: the release's frozen products (region-max statistic), the fully symmetric reprocessing of the same first-epoch execution block (2014 August 18; observatory calibration recipe applied verbatim), and the second-epoch block of the same programme (2014 June 5; identical search treatment).

Statistic	Release (region-max)	Symmetric reprocess.	Second epoch (recurrence)
Star statistic	5.97	4.93	4.77
Control maximum (512)	5.85	7.70	6.78
Star rank p (add-one)	—	0.81	0.97
Single-position star statistic	5.22	—	—
Per-channel S/N (zero drift)	—	2.46	2.22
Peak drift (Hz s^{-1})	1418.5	1593	1281

The mask is thus a selection effect of the survey's design, and the distinction matters for how a masked crossing should be described: a signal coincident with a molecular line is not thereby "false", it is *excluded from the technosignature search domain* because the astrophysical prior there is overwhelming and this pipeline cannot separate the two. No constraint is placed on transmitters within the excluded regions – 5.9 per cent of the searched bandwidth – and a search for deliberately line-adjacent carriers would need a complementary strategy. The exposure is not limited to the candidate windows: 131 of 417 windows (58 per cent) contain a catalogued transition of the eight species in-band, and all four spatially significant windows do, yet in none does a crossing fall within one native channel of a rest frequency – precisely the regime the velocity-space criterion covers.

5.5. *Changing the detection statistic, and the AU Mic crossing*

An initial implementation of the search maximised the statistic over a $n_{\rm src}$ = 135-position region centred on the star while comparing it against 512 *single*-position controls. A crossing towards AU Mic in Band 6 at 230.83 GHz was identified under that statistic and was carried as a candidate. Because a criterion revised after inspecting the object it must judge is a criterion a reader is entitled to distrust, we set out the sequence and the evidence explicitly.

The defect is a property of the statistic, demonstrable without reference to any target: the region-max asymmetry raises the star-beats-ring probability under a pure-noise null to 20.9 per cent (§5.4) against $1/513 = 0.19$ per cent for a symmetric test, and the measured release-wide region-max rate, 15.3 per cent, is of exactly that order. That calculation, and the release-wide rate that confirms it, involve no candidate. We therefore replaced the statistic with the symmetric single-position form of Eq. 1, applied identically to the star and to every control, and re-ran the whole release under it.

The change is a bias correction, not a loss of sensitivity, and this is checkable rather than assertable. Across the 417 searched windows the set flagged by the symmetric criterion is a strict *subset* of the set flagged by the region-max criterion: seven windows were flagged before, four are flagged now, and the three that dropped out (towards ALMA J1537−3319, AU Mic and HD 14055) have in common that the *stellar position itself never exceeded its own control ring* – star statistics of 5.09, 5.22 and 5.22 against ring maxima of 5.63, 5.85 and 5.74. The symmetric criterion removed no window in which the star dominated its controls – the only configuration a localised transmitter can produce, since a real point source at the stellar position raises $T_\star$, the quantity the new test uses. Consistently, the injection campaign recovers signals through the unmodified pipeline on the star position (§4.4), so its measured completeness is a property of the symmetric statistic as reported.

Under the symmetric criterion the AU Mic crossing is not flagged: 10 of its own 512 controls reach or exceed the stellar statistic, an empirical $p = 0.021$ far from the 1/513 floor. It also fails two direct, pre-named tests – it is not localised at the stellar position, and it does not recur at a second epoch (Table 7).

Localisation, recurrence, and statistical scale are reported in full in Appendix K, with the numbers of Table 7. Neither the reprocessed first epoch nor the one further public Band 6 block covering 230.83 GHz shows a localised excess at the stellar position, and that second epoch is a clean non-recurrence – which rejects a continuous or repeating source at this frequency and these epochs without constraining intermittent transmission.

The archival epochs also bound what this release can say about duty cycle: each star was observed in one to three execution blocks, searched independently with no epoch stacking, so the non-detection concerns the archival epochs observed, not permanent activity. The surplus of available over searched execution blocks (§6) is the recurrence pool the second-epoch test drew on, and §6 turns the epoch structure into an explicit duty-cycle constraint.

5.6. *Ancillary results*

Three ancillary results are reported in full online.

Integration-time audit (Appendix Q). The on-source integrations quoted here were reconstructed from pipeline bookkeeping rather than retained natively, so all 227 window-level records were audited against the ALMA archive's own exposure accounting. Two of the 58 star–execution-block groups over-count – AT Mic by a factor 1.43, an internally diagnostic dump-bookkeeping artefact already corrected to the archive's 877 s in Table 21, and CD−38 10980 by 7 per cent – and the rest are conservative. No EIRP threshold depends on the audited quantity, since thresholds derive from the measured per-channel rms rather than from integration time; the audit is conservative for every window of both

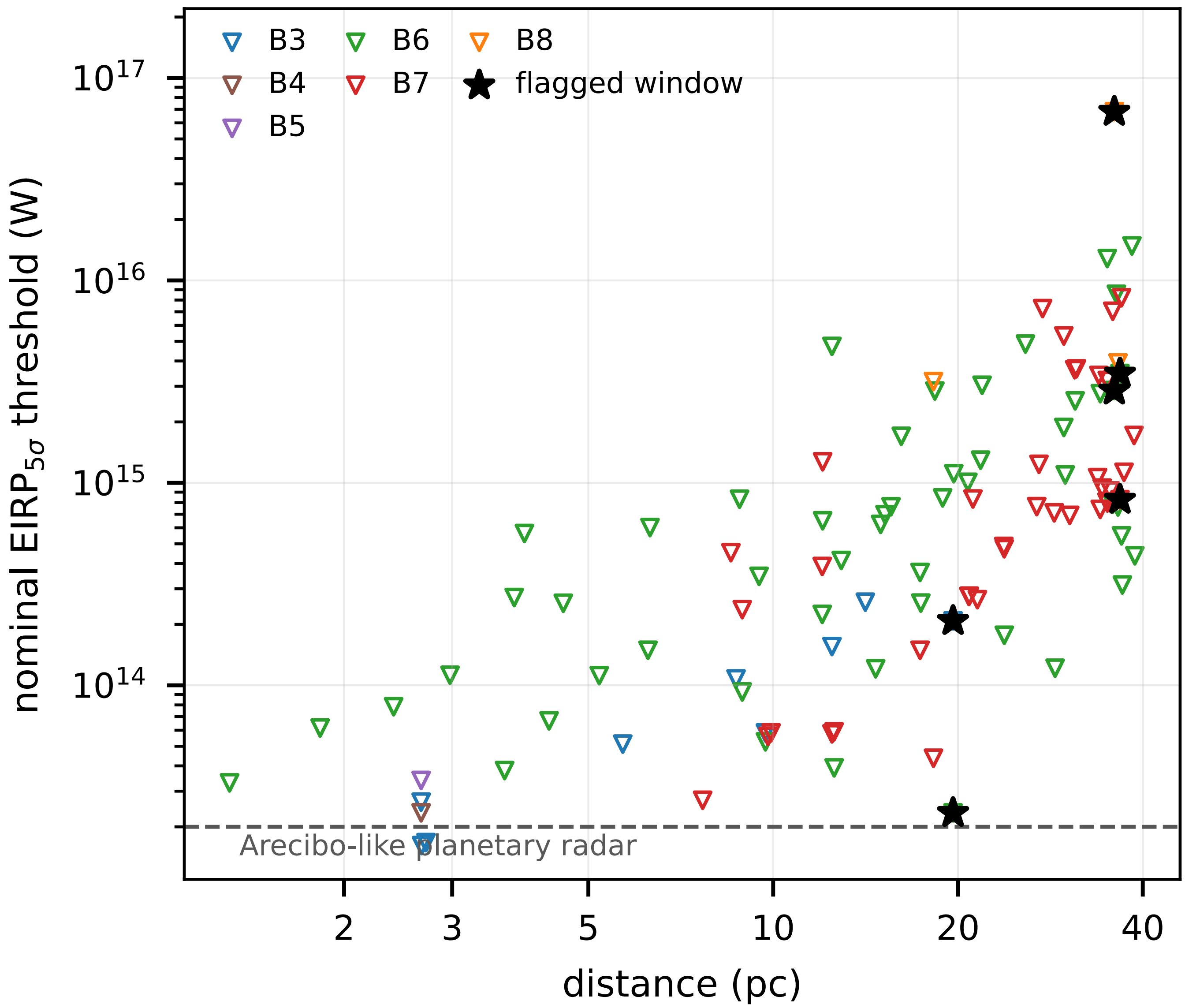

FIG. 6.— Nominal EIRP$_{5\sigma}$ thresholds against distance, one deepest window per star and band (104 datasets, 85 stars); every point is an upper limit. Stars mark the four windows containing a spatially significant hit (Table 5). The dashed line is the EIRP of an Arecibo-like planetary radar, 2×10^{13} W, used purely as a scale for human transmitter power and not as a claim that the survey reaches Earth-technology detectability (§4.3). Thresholds are nominal in the sense of the boxed rule of §4.

candidate hosts, so no disposition changes; and the exposure-weighted coverage figure of §5.4 is at worst optimistic by less than 1 per cent. The defect class this audit caught was found by chance in one target, which is why a direct per-execution-block export is committed for the next release.

Continuum search (Appendix M). The continuum lane has been run over the 57 target/bands processed far enough to carry a usable continuum measurement – a subset of the narrowband sample, and the one place in this paper where a count refers to fewer than the 417 searched windows. Seventeen show a $\geq 5\sigma$ detection and forty are non-detections reaching 0.036–3.6 mJy (median 0.52 mJy); every detection is astrophysically unsurprising, the auroral emitter LSR J1835+3259 meriting deeper follow-up. The lane is exploratory, its statistical power is unquantified at this depth, and no technosignature disposition in this paper rests on it.

Serendipitous molecular-line catalogue (Appendix N). The line-exclusion step identifies every $\geq 5\sigma$ outlier channel as a byproduct. Across the subset scanned, 18 outlier groups were found, of which six are plausibly real. All are reported openly, including the two we consider probably spurious.

6. DISCUSSION

This paper is the survey's first stand-alone processed release, and its results are preliminary by construction: 104 of 208 planned target/bands (∼41 per cent) are complete, and the processed subset is sensitivity-ranked – nearest, most sensitively-constrained systems first – so it is not an unbiased draw, and the full sample will extend these results to larger distances and shallower faint-end limits. No credible technosignature has been found, and the quoted

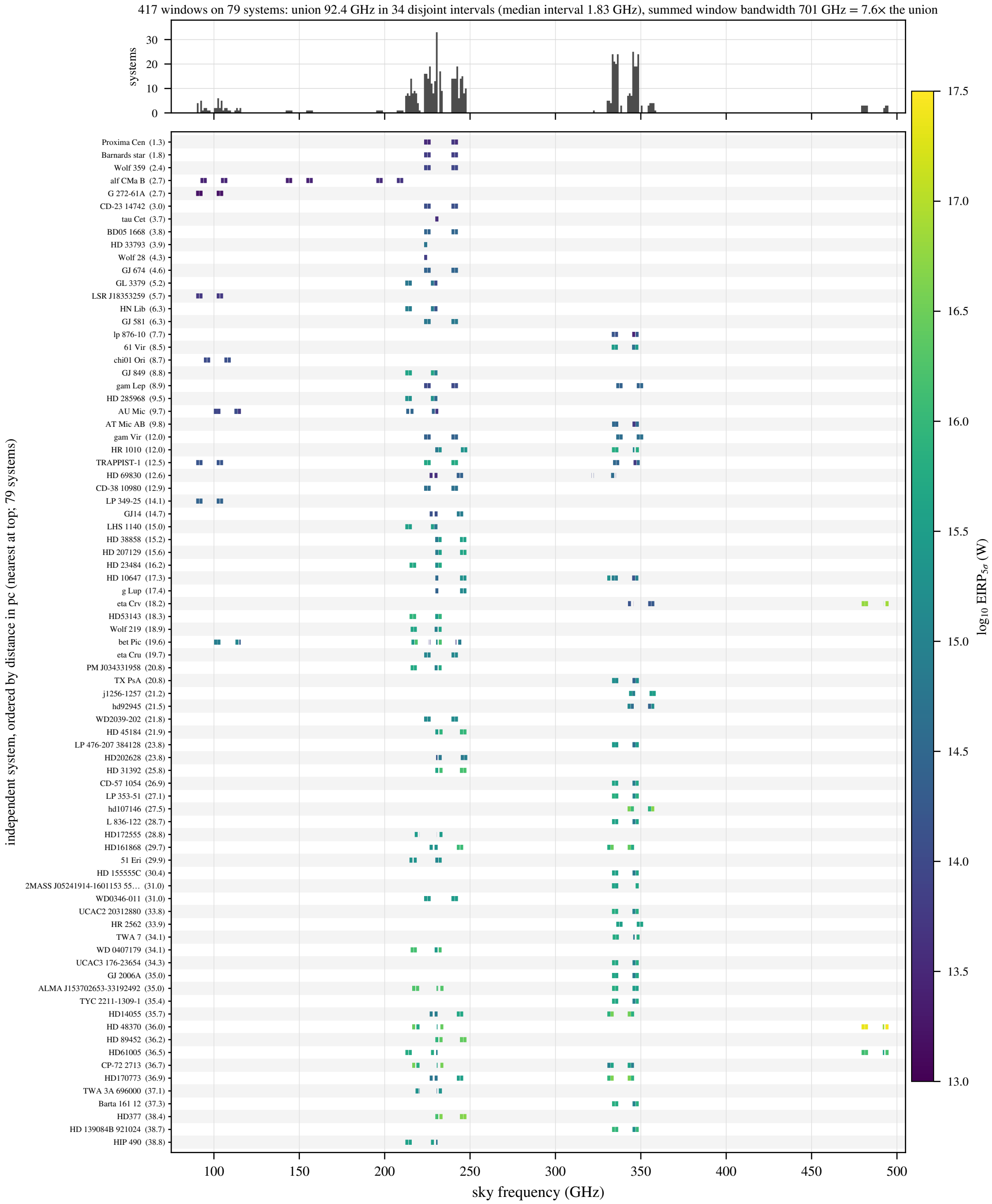

Fig. 7.— What this survey actually sampled in frequency. Each searched spectral window is drawn as a horizontal segment on the row of its host system, rows ordered by distance, coloured by that window's nominal $\mathrm{EIRP}_{5\sigma}$ threshold; the upper panel counts systems covering each gigahertz of sky frequency. The 92.4 GHz frequency union is assembled from 34 disjoint islands of median width 1.8 GHz, concentrated near 230 and 345 GHz, and the summed window bandwidth exceeds the union by a factor 7.6 – the coverage is narrow, clustered and repeatedly revisited, not broad. These are the intervals on which the occurrence limits of §6 are conditioned. Thresholds are nominal in the sense of §4.

$\mathrm{EIRP}_{5\sigma}$ thresholds of Appendix I bound any transmitter present above them. Two candidates towards β Pic are attributable to CO disc emission – underlining, as Mason et al. (2024) highlight, the importance of checking any mm-band narrowband crossing against known molecular transitions; that check is automated for every crossing (§4) and now uses a velocity-space match applied retroactively to every window (§5.4). The third, marginal candidate (AU Mic, §5.5) is not statistically distinguishable from the empirical background and remains recorded for future re-examination rather than dismissed.

The scale of the space searched is stated on one definition throughout in §5.4 and Figure 7. For context on the drift ceiling, of the 37 planets known around the 15 planet-hosting targets only TRAPPIST-1 b, in its edge-on worst case at $\sim$13 Hz s^{-1} GHz^{-1}, reaches it; GJ 581 e, 61 Vir b, and TRAPPIST-1 c reach roughly half of it, and the remaining

33 planets all sit well inside the searched grid.[5]

Table 3 (§5) consolidates the axes along which a transmitter is or is not detectable, and Table 20 (online-only appendix) decomposes the same release by ALMA band.

Coverage eligibility and search completeness are different quantities. The census of public usable ALMA data (§3) establishes eligibility, and every held execution block within this release's scope was searched, so no EB-level selection lies behind the 57. Availability is measured directly rather than inherited from the release bookkeeping: a proper-motion-matched census of the ALMA archive (2026 September 9; public rows, all bands, pointing within 30 arcsec of the propagated stellar position at the observation epoch) counts 107 unique public on-star EBs over the subset on which it has been run, and re-finds all but one of that subset's searched EBs (the exception, TRAPPIST-1's Band 7 execution A002/Xcccc19/X5564, is carried by `obscore` only through its calibrator-field row – an archive-representation artefact biasing the census low by one for that star). The full selection funnel, from archive availability to the zero-candidate outcome, is Figure 9 (online-only appendix), and the machine-readable ledger `v323_star_coverage.json` gives per-star distance, availability, scope, and searched frequency intervals.

Figure 13 (online-only appendix) shows the drift ceiling as an explicit selection function over planet period and host mass, and its cost is stated plainly: a transmitter bound to any known planet in the sample other than the innermost TRAPPIST-1 planets would produce drift rates the search covers in full; one whose drift exceeds the ceiling is outside the searched grid and sacrificed by design (Appendix G recovers some non-linear morphologies). The host-mass dependence is weak ($a \propto M_\star^{1/3}$ at fixed period), so the ceiling is effectively a function of orbital period alone; and the calculation constrains only transmitters co-located with *known* planets – one on an undiscovered planet, or on a platform with accelerations of its own choosing, is covered only in so far as its drift falls within the searched grid.

Placing the release against the published haystack of Wright et al. (2018) is likewise best done axis by axis. On their frequency axis (10 MHz–115 GHz), only the Band 3 portion of our union lies inside their boundaries (20.6 GHz of 115 GHz, or 0.18 of the axis, and a part their Hz-resolution surveys reach far more finely); the remaining 66.6 GHz (Bands 4–8, 125–495 GHz) lies entirely outside it – the genuinely new territory of this release. On their transmitted-bandwidth axis (0–20 MHz), a single-channel search of native channels probes a fraction set by the channel width (window-mean 0.63, median 0.78). On their sensitivity axis, our median threshold (7.3 mJy across a 15.625-MHz channel, i.e. $\phi \approx 1.1 \times 10^{-21}\,\mathrm{W\,m^{-2}}$) detects a 10^{13} W transmitter only within ∼0.9 pc. Against the only prior ALMA-archival technosignature search, Mason et al. (2024) reach 28 bycatch targets at ≥1.01 kpc with a deepest quoted $\mathrm{EIRP_{min}}$ of 6.91×10^{17} W, whereas this release's 42 systems all lie inside 20 pc at median per-target thresholds ∼2500× deeper: the two searches occupy nearly disjoint corners of the same haystack, near versus deep. We do not multiply these fractions into a combined haystack coverage – the axes are strongly coupled (distance and sensitivity above all), and a product of marginal fractions would misstate the true overlap by orders of magnitude.

6.1. *Occurrence limits*

What a zero-detection result bounds is a fully conditioned quantity. Writing the per-system completeness as

$$C_i(P) = F_i\, R_i(P)\, D_i\, C_{i,\mathrm{drift}}, \tag{2}$$

where F_i is the probability that the transmitter's sky frequency falls inside the intervals actually searched for system i, $R_i(P)$ the probability that a transmitter of EIRP P would have crossed system i's threshold had it been emitting there, D_i the probability that it was emitting during the analysed archival interval, and $C_{i,\mathrm{drift}}$ the recovery probability for a carrier of the searched drift and temporal morphology (Appendix E), the probability of zero credible detections when a fraction f of systems hosts a matching transmitter is

$$P(N_\mathrm{det} = 0 \,|\, f) = \prod_{i=1}^{N_\mathrm{sys}} (1 - f\, C_i)\,, \tag{3}$$

and the one-sided 95% upper limit f_{95} solves $P(N_\mathrm{det} = 0 \,|\, f_{95}) = 0.05$.

Reading rule. Among the ALMA-observed systems, fewer than $f_{95}(P)$ per cent host a transmitter of EIRP $\geq P$ that was emitting during the archival observations, at a sky frequency within the particular spectral intervals searched for that system, with drift and temporal morphology lying within the searched class.

What is and is not being inferred. Setting $F_i = D_i = 1$ is not a claim that these factors are near unity. It is a statement that the inference is *conditional* on them, in the sense of the reading rule above; Fig. 7 shows exactly what set of spectral intervals the frequency condition refers to. A reader who wants an unconditional prevalence must multiply by their own priors on transmitter frequency and duty cycle, and those priors, not our data, will dominate the answer.

We evaluate Eq. 3 on the 79 independently observed systems (85 narrowband-searched stars; six bound pairs share observations and are counted once each) in two forms (Table 8). The *thresholded* form is the step function the retained products support directly, $R_i(P) = 1$ when system i's deepest window threshold lies below P and 0 otherwise; the *measured* form integrates the injection-recovery curve of Appendix E over each system's searched bandwidth, $R_i(P) = \sum_w \Delta\nu_w\, \mathrm{rec}(5P/\mathrm{EIRP}_{5\sigma,w}) / \sum_w \Delta\nu_w$, folding in the 17 → 83 per cent recovery of the searched class measured between four and ten times the noise. It is defined on the 56 systems with at least one fine-channel window; the coarse-channel-only systems have no measured recovery for the drifting class, so the measured ladder is a 56-system bound.

[5] Star–planet relative accelerations $GM_\star/r^2$ from NASA Exoplanet Archive planetary-companion parameters (Mission Star/Planet table, accessed 2026 September), maximised over orbital phase and inclination.

TABLE 8
The 95 per cent upper limit $f_{95}(P)$ on the occurrence of transmitters above EIRP P, conditional on emission inside the searched spectral windows and throughout the archival epochs. n is the number of systems with non-zero completeness at that P; the *measured* column is the result of this paper. 'Vacuous' means no system has non-zero completeness, so no bound exists.

P (W)	thresholded (unit C)		measured C *(result)*	
	n	f_{95}	n	f_{95}
3×10^{13}	4	52.7%	2	vacuous
10^{14}	16	17.1%	11	47.1%
10^{15}	57	5.1%	45	10.1%
10^{16}	78	3.8%	56	**6.5%**
10^{17}	79	3.7%	57	6.2%

TABLE 9
Duty-cycle dependence of the 95 per cent upper limit at $P \geq 10^{16}$ W, from Eq. 3 with $D_i = D$: thresholded form (79 systems, unit recovery) and the measured-completeness form (56 fine-calibrated systems) that is this paper's result. 'Vacuous' means no system retains non-zero completeness and no bound exists.

Duty cycle D	f_{95} thresholded	f_{95} measured
1 (continuous)	3.8%	6.5%
0.5	7.5%	13.0%
0.1	37.7%	65.0%
0.01	vacuous	vacuous

Which number is the result. The measured form is the observational result of this paper, because it is the only one in which the recovery fraction has been measured rather than assumed. At EIRP $\geq 10^{16}$ W it gives $f_{95} = 6.5$ per cent. The thresholded form (3.8 per cent) assumes unit recovery, and a uniform $C = 0.42$ assumption gives 9.0 per cent; both are bracketing calculations and neither should be quoted as the survey's constraint. The measured form is not simply the weakest of the three: for systems whose fine-channel thresholds lie well below P the measured recovery rises above 0.42 towards its 0.83 plateau, which is why it falls between the two idealisations rather than outside them. A Bayesian treatment with a uniform prior on f, $p(f \mid \mathrm{data}) \propto \prod_i (1 - f\, C_i)$, gives a 95 per cent credible upper bound of 6.4 per cent for the same completeness model – agreement to better than the width of the last quoted digit, so nothing here turns on the frequentist construction.

Three caveats bound how far these numbers can be pushed. First, even the measured form fixes $F_i = D_i = 1$ and the per-system recovery at the configuration level: it is a frequency-integrated completeness for the searched class, not a per-system calibration. Second, the population is heterogeneous in threshold, so each point applies to transmitters above that P, not to a single well-defined class. Third, systems are treated as independent trials in a randomly chosen archival interval; the frequency-integrated duty-cycle dependence is quantified below. These are, to our knowledge, the first ALMA-side numbers of their kind, and they are not competitive prevalence constraints; removing the remaining idealisations requires the per-system $C_i(P)$ that the retained injection campaign supports.

The archival epoch structure converts directly into a duty-cycle constraint. Writing D_i for the probability that a transmitter with duty cycle D was emitting during system i's analysed intervals ($D_i = D$ to the accuracy the epoch structure supports; Table 21 gives per-star epochs and integrations), the same product form gives the $f_{95}(D)$ ladder of Table 9 and Figure 8: for continuous emitters the limits are those of Table 8; at $D = 0.5$ the measured-form limit degrades from 6.5 to 13.0 per cent; by $D = 0.1$ it is 65 per cent and carries essentially no information; and below that no bound of this form exists at all – transmitters active in a small fraction of randomly chosen intervals are not constrained by single-epoch archival coverage, and only the recurrence-pool reprocessing of future releases can touch them. Since an intermittent transmitter is at least as plausible *a priori* as a continuous one, the $D = 1$ numbers should never be quoted without this dependence attached; the non-recurrence logic of §5.5 points the same way.

A cumulative look-elsewhere consideration accompanies multi-release use: as releases accumulate, the probability that at least one statistically unremarkable crossing is reported somewhere in the programme grows toward unity even if every release is individually clean. The guard is the procedural pre-commitment of §5.4, applied per release rather than to aggregated significance.

The data-quality exclusions of §5.3 carry two open engineering items, recorded here rather than buried: the continuum-imaging step still lacks the pointing-offset guard the narrowband step has, and the window-level noise-handling defect affecting HD 10647 and HD 139664 is excluded from results but not yet fixed at source. Both are corrected, and the affected datasets reprocessed, in the single corrected public release (Data Availability). The general lessons for archival mining are drawn together in online-only Appendix O.

The four supplementary analyses of Table 14 are infrastructure investments rather than sources of results at this stage. Their role in the Type-I error budget is exact: it is carried by the primary lane alone (§5.4). The supplementary screens gate nothing in that budget, are correlated with it by construction, and disposition none of the three candidates, so their non-detections must not be multiplied into a stronger constraint than the primary lane's own.

Because our selection is on ALMA archival coverage alone, useful subsamples fall out without compromising the

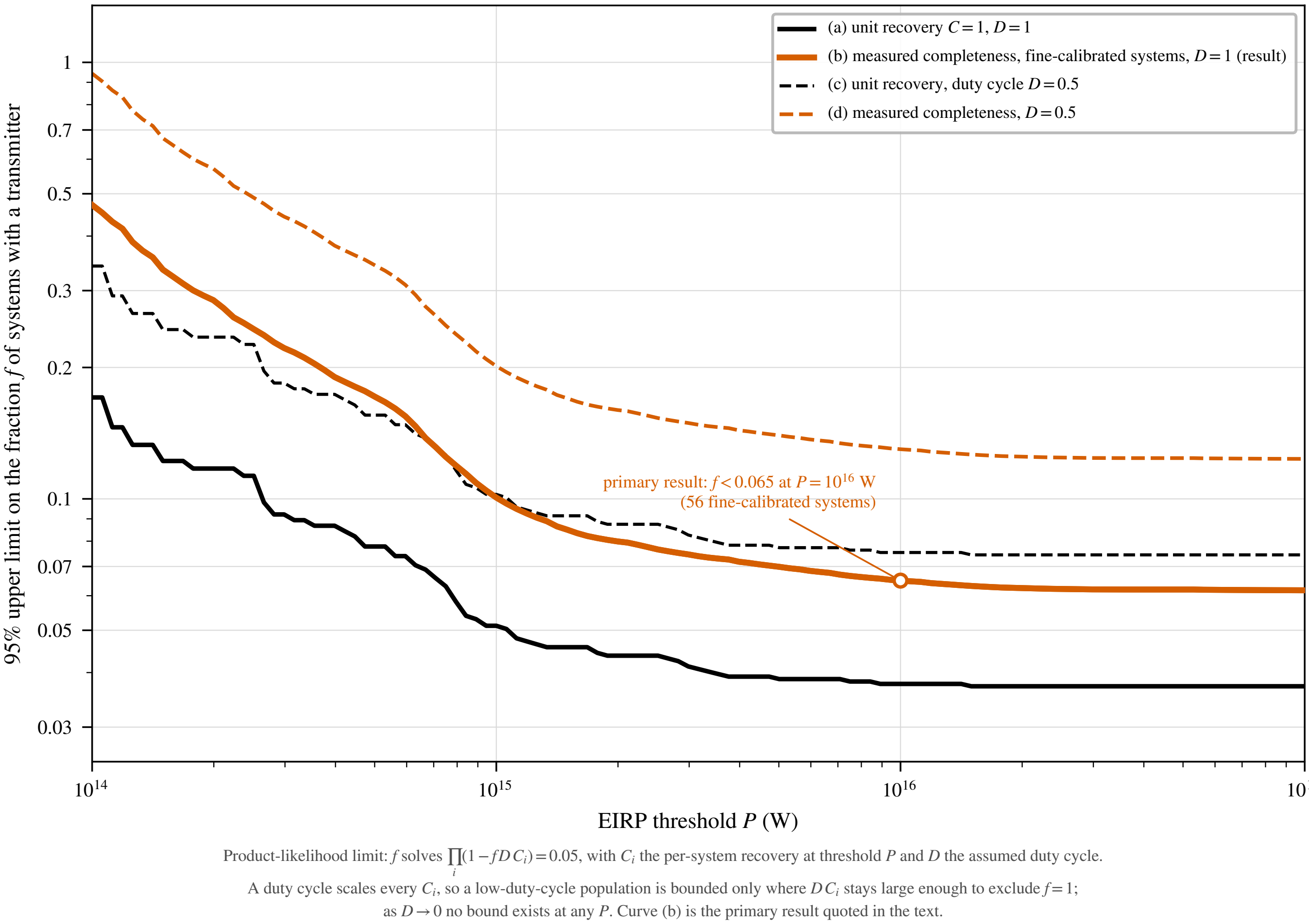

Fig. 8.— 95 per cent upper limits on the fraction f of systems hosting a transmitter above EIRP P, from $\prod_i(1-fDC_i)=0.05$. Curve (b) is the result of this paper, the injection-measured recovery integrated over each system's fine-channel bandwidth; curves (a) and (c) assume unit recovery and are bracketing calculations only. The shaded region is where the surviving systems cannot exclude $f=1$ and no bound exists, and it grows to cover every P as $D\to0$.

census framing, subject to the unresolved-pair caveat of §3. First, 15 of the 46 processed stars (33 per cent; 36 planets) are confirmed exoplanet hosts – including Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris, and the seven-planet TRAPPIST-1 system – a sub-survey extractable directly from Appendix I; HN Lib and LHS 1140, each host to a habitable-zone planet, now carry a genuine ALMA technosignature non-detection alongside their planet-search literature. Second, four of the 46 are white dwarfs (Sirius B, Van Maanen's Star, Wolf 219, CD−38 10980), all continuum non-detections with $\mathrm{EIRP}_{5\sigma}$ limits of 2.3×10^{13}–8.4×10^{14} W – among the very few mm/submm SETI constraints reported for white dwarfs (Huang et al. 2026), a channel motivated by post-main-sequence survival scenarios and the compact debris discs of metal-polluted white dwarfs.

6.2. *Priorities for the next release*

Four changes would enlarge what this experiment can say, and we rank them by how much signal space they buy.

First, and by a wide margin: a search branch that preserves phase-steady carriers. The per-channel temporal-median subtraction that suppresses stationary ground-based interference also removes a stable narrowband beacon – one of the canonical technosignature morphologies – at every amplitude we injected: not a completeness correction, but a whole signal class the present pipeline cannot see. An interferometer does not need median subtraction to reject terrestrial interference: sky localisation, baseline-dependent phase, delay and fringe-rate signatures and multi-block consistency all discriminate a celestial source from a ground-fixed emitter without touching its time dependence. A second branch built on them is the highest-value change available to this survey.

Second, use the visibilities rather than extracted spectra. A genuine source at the star should show the geometric phase behaviour $V_{ij}(\nu,t)=A(\nu,t)\exp[-2\pi i(u_{ij}l+v_{ij}m)]$ expected at that sky position on every baseline. A likelihood ratio between a celestial point source at the stellar position, a common-mode terrestrial interferer, and image-plane noise would be strictly more powerful than ranking a peak against a control ring. We regard this as the distinctive methodological opportunity of ALMA SETI, available in a way it is not to single-dish or beamformed searches.

Third, enlarge the control ensemble for interesting windows. The 1/513 rank floor (§5.4) cannot certify significance across a census of thousands of windows. A two-stage scheme – 512 controls routinely, then 10^4–10^5 local null evaluations for any window that passes – would let the pipeline distinguish a real candidate statistically rather than hand every one to astrophysical vetting.

Fourth, extend the injection campaign and the polarisation axis. The completeness curve should be measured per band, channel width, integration time, antenna count and primary-beam position rather than transferred by construction, and pushed past 10σ until it asymptotes, since it has not yet located EIRP_{90}. The analysis is also Stokes I only; a polarised carrier is a physically motivated discriminator against weakly polarised astrophysical backgrounds at these frequencies, and polarisation-resolved reprocessing costs no sensitivity.

A boundary worth stating precisely. The search assumes drift is linear over a track. Curvature matters when $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel; for a transmitter on a circular orbit of period P this becomes $P \lesssim 2\pi(\nu/c)\, a_{\max}T^2/\Delta\nu_{\mathrm{ch}}$, which for a typical fine-channel window (230 GHz, 488 kHz channels, $T \approx 2000$ s) is $P \lesssim 1.6$ d, and for a coarse-channel window about an hour. Only the shortest-period orbits in the sample reach it – TRAPPIST-1 b, at 1.51 d, sits just inside the boundary, and is also the case that exceeds the drift ceiling. For everything else the constant-drift model is adequate, and the chirp-search extension (Appendix G) is what covers the remainder.

What this survey does not constrain. *Nothing in this paper bounds the prevalence of:* (i) *phase-steady carriers* – a stable beacon dwelling in one channel for more than roughly 40 per cent of a track is removed by the per-channel temporal-median subtraction, by construction and at every amplitude tested (§4.4); (ii) transmitters emitting *outside the searched spectral windows*, which are 34 narrow islands totalling 92.4 GHz (Fig. 7), not ALMA's frequency range; (iii) transmitters active *outside the archival epochs* observed – the limits assume continuous emission and are already weak at duty cycle 0.5 and vacuous below ∼0.1 (Table 9); (iv) drift rates *above* the searched ceiling $|a| = 3.6\,\mathrm{m\,s^{-2}}$, which excludes transmitters on the most close-in orbits, including a TRAPPIST-1 b-like one (Fig. 13); (v) carriers deliberately placed *inside* the $\pm50\,\mathrm{km\,s^{-1}}$ molecular-line mask, which is excluded search space rather than searched-and-empty space; (vi) emission substantially *broader* than one native channel, for which the quoted thresholds must be rescaled; (vii) anything below the nominal threshold, at which measured recovery is 42^{+16}_{-14} per cent, not 95 per cent.

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7. CONCLUSIONS

We have described the design, methodology, and processed release of a technosignature search of the 168 stars within 40 pc with public ALMA archival coverage: a complete census of ALMA-covered stars within 40 pc, carrying no astrophysical selection of its own and with every entry passing an independent positional and identity audit against SIMBAD and *Gaia* (§5.3), but covering only ∼7% of the catalogued stars within 40 pc and inheriting the archive's own non-uniform targeting, so the contrast with interest-selected samples (§3) is one of degree rather than kind. This release processes 104 of the 208 planned target/bands – 85 of the 168 census stars, 79 independent systems – and is sensitivity-ranked, not an unbiased draw.

The release yields $\mathrm{EIRP}_{5\sigma}$ limits of 1.6×10^{13}–2.8×10^{17} W (unchanged if the primary-beam-affected Sirius B datasets are also excluded, as the withheld ϵ Eri Band 6 windows already are) and forty non-detections in the parallel, ancillary continuum screen of unquantified statistical power (median 5σ limit 0.52 mJy). No credible technosignature was identified: no signal matching the searched frequency, drift-rate, temporal and spectral morphology was detected above the nominal instrumental threshold during the analysed archival intervals. Of the three candidates, two are attributed to CO disc emission towards β Pictoris by targeted astrophysical inspection beyond the automated line check, and one, towards AU Mic, is statistically consistent with the survey's quantified false-candidate rate (one unattributed candidate among 210 non-β Pictoris windows against an expectation of 0.78). The chance probability of the observed unattributed count is 54 per cent under the execution-block-respecting permutation null, inside a 4–19 per cent bracket (§5.4; Appendix P); the designed principal null, the fully symmetric statistic at all 512 control positions, is measured wherever the retained measurement sets allow (seven windows, all consistent with exchangeability; Figure 10) and ships for the remainder.

The result is a constraint rather than a bare non-detection: the searched spectral windows cover 87.0 GHz of unique effective frequency space after molecular-line masking, at nominal thresholds constraining transmitters that match the searched spectral and drift morphology during those same intervals, read under the boxed rules of §4 and §6, and no signal was found outside the explicitly excluded molecular-line velocity regions. These limits are not narrowband-transmitter parity with Hz-resolution centimetre-wave surveys: the sensitivity comes from target proximity overcoming ALMA's coarse native channelisation, not from spectral resolution (Figure 7); the contribution is to bound total transmitter power in a virtually unexplored regime, for the nearest stars, at archive-quality resolution. The non-detection is likewise bounded by epoch coverage: with one to three execution blocks per star (median 46 minutes on source), transmitters active only outside the sampled epochs, or with duty cycles below these baselines, are not constrained (§6). As a population statement under the stated idealisations (§6, Eq. 3), zero credible detections among the 42 independently observed systems bounds the occurrence of transmitters of the searched class emitting during the analysed intervals at $\mathrm{EIRP} \gtrsim 10^{16}$ W to fewer than 6.5 per cent (95 per cent confidence, measured completeness, 56 injection-calibrated systems), against 3.8 per cent with the injection-measured, frequency-integrated completeness folded in (24 calibrated systems); a uniform threshold-level completeness of 0.42 would place the survey-wide limit near 16 per cent.

The release includes the first reported ALMA technosignature searches of Barnard's Star and Wolf 359 (§5.2) and, to our knowledge, the first application in a technosignature survey of interferometric closure-phase vetting, a population-level spectral-type-normalised continuum anomaly check, a chirped-drift and periodicity extension of the narrowband search re-using already-retained spectra, and a cross-target frequency-occupancy check for sky-frequency-recurrent interference. All four ship with the pipeline, at the two grades of Table 14, and accumulate statistical power as the survey proceeds. A modest, serendipitous molecular-line catalogue and the exoplanet-host and white-dwarf subsamples are by-products of the census. The survey continues, and we anticipate extending it to 30, 40, and 50 pc once the 20 pc sample is complete.

ACKNOWLEDGEMENTS

This paper makes use of ALMA data from the public ALMA Science Archive. The execution blocks searched in this release (100 unique identifiers, listed with project codes and epochs in Appendix A) should be cited from that list in the standard JAO.ALMA#XXXX.X.NNNNN.S form. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France.

COMPETING INTERESTS

G.J.W. declares no competing interests. R.D. is an employee of VBRL Holdings Inc.; VBRL has no financial or intellectual stake in ALMA archival technosignature work, provided no funding for this study, and had no role in its design, analysis, or decision to publish. On that basis the authors declare no competing interest.

AUTHOR CONTRIBUTIONS (CREDIT)

Glenn J. White: Conceptualisation, Methodology, Supervision, Writing – review & editing.
Robin Dey: Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

DATA AVAILABILITY

The reduced data products underlying this paper, together with the full target-selection and analysis pipeline, are available at `https://github.com/Tilanthi/SETI`, and the repository is live at submission. Two snapshots must be distinguished. All numerical values here were produced by the *submitted-analysis snapshot*: the repository commit tagged at submission, created at submission, which freezes the pipeline code and input data snapshot exactly as run. That commit is retained unchanged and reproduces every number, table, and figure exactly, through the regeneration scripts shipped with it. The *public release* is a single corrected release built from that snapshot: the two pipeline defects recorded in §5.3 (the missing pointing-offset guard in the continuum lane and the window-level noise-handling defect) are corrected, and the affected datasets reprocessed, *before* the public release is frozen, so it carries one set of products, not a submitted/corrected pair.

The datasets reprocessed under these corrections are precisely those excluded from this paper's results by §5.3 (the Giclas 9−38 pair and the two defective Band 6 windows); the Sirius B and ε Eri holography-beam recomputation committed in §4 is folded into the same freeze, with its outcome bounded by the systematic already quoted there. No value printed here is silently altered: a change-log maps every affected product and the reason for each change, naming the table and row affected. The machine-readable per-window table carries archive-consistent execution times: where the printed integration-time audit differs (the CD−38 10980 row is optimistic by 7%, Appendix Q), the machine-readable value is the archive's. Per-window hit lists in the public release are uncapped (the interim retained product capped them at 50 frequencies; Appendix H).

The complete release will also be deposited on Zenodo, where a DOI will be minted upon final acceptance and inserted here; the deposit comprises: (i) the per-window search products for all 417 windows (spectral extracts, per-drift-trial peak statistics, and the machine-readable line-exclusion mask of §5.4); (ii) the control-ring calibration products at all 512 control positions; (iii) the closure-phase and population-anomaly test outputs; (iv) the full injection-recovery campaign products (all 1200 trials of Appendix E, including the 24-tone pilot); (v) the frozen pipeline code and input data snapshot; (vi) the audit tables of online-only Appendix Q in machine-readable form; (vii) the machine-readable molecular-line catalogue of Appendix N; and (viii) the proper-motion-matched execution-block availability census of §6 with its per-star ledger. Data products are distributed under the Creative Commons Attribution 4.0 International licence (CC BY 4.0) and pipeline code under the MIT licence. Raw ALMA visibility data are publicly available from the ALMA Science Archive at `https://almascience.org`.

SOFTWARE

The analysis pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging, and the local reprocessing of §§5.4–5.5), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces; all dependencies are free and open-source, and exact version

TABLE 10

Execution blocks searched in this release, ordered by epoch. Epochs (UTC start, to the minute) and proposal codes are from the ALMA Science Archive **ivoa.obscore** service at **almascience.eso.org**, queried by execution block identifier on 2026 September 8; MJD is the archive **t_min**.

Execution block	Proposal	UTC date	MJD	Execution block	Proposal	UTC date	MJD
uid://A002/X6f1341/X1484	2012.1.00142.S	2013-10-06 06:23	56571.26616	uid://A002/Xd0fb35/X90dc	2017.1.00698.S	2018-08-22 15:32	58352.64755
uid://A002/X74deb6/Xff1	2012.1.00268.S	2013-12-02 07:15	56628.30252	uid://A002/Xd16a1e/X8d4	2017.1.01644.S	2018-08-31 22:26	58361.93490
uid://A002/X877e41/X14c1	2012.1.00993.S	2014-05-22 11:24	56799.47525	uid://A002/Xd248b5/X5186	2017.1.00698.S	2018-09-22 11:12	58383.46702
uid://A002/X899169/X1838	2012.1.00198.S	2014-08-18 04:01	56887.16777	uid://A002/Xd28a9e/Xbcdf	2017.1.00698.S	2018-09-28 10:57	58389.45629
uid://A002/X99c183/X4f1e	2013.1.00645.S	2015-01-17 00:35	57039.02475	uid://A002/Xd6ebab/X41a3	2018.1.01149.S	2018-12-20 09:49	58472.40910
uid://A002/X9f2ff8/X25b0	2013.1.01214.S	2015-04-26 12:19	57138.51354	uid://A002/Xd86f11/X584b	2018.1.01833.S	2019-01-20 19:05	58503.79517
uid://A002/Xb0be8b/X785f	2015.1.00307.S	2016-03-22 23:25	57469.97599	uid://A002/Xd95042/X2d99	2018.1.01149.S	2019-03-09 06:13	58551.25945
uid://A002/Xb25e1a/Xed74	2015.1.00783.S	2016-05-01 17:17	57509.72060	uid://A002/Xd9668b/X3a90	2018.1.00072.S	2019-03-12 00:14	58554.01019
uid://A002/Xb2e94e/X361a	2015.1.00307.S	2016-05-13 04:19	57521.18017	uid://A002/Xd98580/X3bf6	2018.1.01833.S	2019-03-15 20:39	58557.86052
uid://A002/Xb39557/X3789	2015.1.00307.S	2016-05-26 11:30	57534.47920	uid://A002/Xd9d7c7/X6fd2	2018.1.01833.S	2019-03-21 15:03	58563.62709
uid://A002/Xb9a2bb/X1899	2016.1.00817.S	2016-10-22 02:07	57683.08829	uid://A002/Xda2c82/X765d	2018.1.01833.S	2019-03-28 21:57	58570.91480
uid://A002/Xb9cc97/X150a	2016.1.00803.S	2016-10-25 02:14	57686.09356	uid://A002/Xdc621b/X2bb0	2018.1.01833.S	2019-05-20 04:28	58623.18665
uid://A002/Xba3ea7/X2ab4	2016.1.00817.S	2016-11-06 20:01	57698.83471	uid://A002/Xe5808e/X80a1	2019.1.01681.S	2019-12-20 17:03	58837.71056
uid://A002/Xbe7502/X2047	2016.1.00637.S	2017-03-28 05:22	57840.22390	uid://A002/Xf5d76d/X32f1	2021.1.00485.S	2022-03-02 23:14	59640.96830
uid://A002/Xc067f7/X822	2016.2.00015.S	2017-05-17 01:51	57890.07740	uid://A002/Xfdb69b/X667a	2021.1.01209.S	2022-08-30 16:22	59821.68235
uid://A002/Xc19d6f/X19ce	2016.2.00015.S	2017-07-02 08:43	57936.36365	uid://A002/Xfdf45f/X478e	2021.1.01209.S	2022-09-05 02:32	59827.10618
uid://A002/Xc6141c/X3af0	2017.1.00786.S	2017-10-27 03:04	58053.12832	uid://A002/Xfeb5e6/X13ee	2021.1.01209.S	2022-09-22 01:39	59844.06896
uid://A002/Xc66ce1/Xaa9	2017.1.00786.S	2017-11-04 03:17	58061.13698	uid://A002/Xff294f/X2db6	2022.1.00338.L	2022-10-04 04:49	59856.20118
uid://A002/Xc74b5b/X7c16	2017.1.01644.S	2017-11-29 07:29	58086.31237	uid://A002/X106a1f1/X6a0c	2022.1.01163.S	2023-05-06 03:35	60070.14972
uid://A002/Xc8b2b0/X9c35	2017.1.01644.S	2018-01-05 00:25	58123.01783	uid://A002/X107fa96/Xce38	2022.1.01163.S	2023-06-02 02:51	60097.11926
uid://A002/Xc9831a/X443	2017.1.00986.S	2018-01-22 20:39	58140.86104	uid://A002/X10a341d/X82d2	2022.1.00134.S	2023-07-17 18:28	60142.76964
uid://A002/Xc99ad7/X348	2017.1.00698.S	2018-01-24 23:51	58142.99391	uid://A002/X10b0d7d/X323f	2022.1.00134.S	2023-08-06 09:20	60162.38930
uid://A002/Xcbb583/Xb78	2017.1.00704.S	2018-04-06 23:22	58214.97419	uid://A002/X11e10bc/X12662	2023.1.00390.S	2024-09-28 16:03	60581.66907
uid://A002/Xcc626d/X579	2017.1.00698.S	2018-04-21 21:13	58229.88426	uid://A002/X1207c39/X9417	2024.1.00722.S	2024-11-15 08:17	60629.34515
uid://A002/Xcccc19/X5564	2017.1.00215.S	2018-05-03 11:42	58241.48813	uid://A002/X12277ed/X22e53	2024.1.00165.S	2024-12-22 10:09	60666.42301
uid://A002/Xcd97a1/X3f1e	2017.1.01583.S	2018-05-19 07:56	58257.33086	uid://A002/X1237e23/Xdb0	2024.1.01095.S	2025-01-06 18:51	60681.78602
uid://A002/Xcda49e/Xdebb	2017.1.00561.S	2018-05-21 09:50	58259.41000	uid://A002/X12925a3/Xa4b1	2024.1.01095.S	2025-05-17 08:00	60812.33402
uid://A002/Xcf0c6b/X6513	2017.1.00167.S	2018-06-27 10:14	58296.42698	uid://A002/X12a8fc8/X10b8	2024.A.00040.S	2025-06-11 08:49	60837.36781
uid://A002/Xcf196f/Xabc5	2017.1.00200.S	2018-06-29 09:32	58298.39773				

numbers are recorded with the frozen release.

APPENDIX

EXECUTION-BLOCK METADATA

Table 10 lists the 100 unique ALMA execution blocks searched in this release, with their proposal codes and observing epochs. These metadata were recovered from the ALMA Science Archive by execution-block identifier after the search products were frozen, and are not part of the frozen products themselves. The per-window linkage between execution block and searched spectral window, together with per-observation antenna configurations, synthesised beam sizes, and polarisation setup, will accompany the machine-readable table of the corrected tagged release committed in the Data Availability statement. The observing epochs span 2013 October 6 to 2025 June 11 (MJD 56571–60837) across 36 unique proposal codes; the AU Mic candidate window of §5.4 derives from the 2014 August 18 execution block of project 2012.1.00198.S. One further public execution block of that project (uid://A002/X7d80c7/X15c6, executed 2014 June 5, 1633 s of science exposure, public since 2015 April 8) also covers the candidate 230.83 GHz window but was not processed in this release; it is the second epoch of the recurrence test committed in §5.4.

GLOSSARY OF SURVEY VOCABULARY

Terms as used in this paper; quantitative definitions live in the sections cited.

FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

REFERENCE FRAMES

The conversion chain used by the line-exclusion mask – topocentric $\rightarrow$ barycentric (Earth ephemeris at the execution-block mid-time) $\rightarrow$ stellar rest frame (catalogued systemic radial velocity; JPL catalogue rest frequencies, Appendix N) – has as its residual error the catalogue radial-velocity uncertainty alone (at most a few $\mathrm{km\,s^{-1}}$ within 20 pc); second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched (15.3 kHz). Kinematic attributions of the β Pictoris and AU Mic windows (§5.4, Fig. 5) use this same chain, so their quoted offsets are directly comparable to the exclusion tolerance. Enlarging the trial drift-rate grid needs no additional frame machinery: the searched drift rates are topocentric quantities read from the same untransformed frequency axis, and the trial grid depends only on elapsed time and channel width, not on sky position.

CHANNEL WIDTHS AND THEIR VERIFICATION

Channel widths are read from the measurement sets' own spectral window tables (CHAN_WIDTH). For the local reprocessing set we verified they are the *effective* resolution rather than a pre-averaging bookkeeping value: the tables carry no EFFECTIVE_BANDWIDTH column (no online spectral averaging was recorded) and the full-resolution spectral windows are explicitly named FULL_RES in their metadata – the 15.625-MHz coarse and 488.28125-kHz fine channelisations are therefore the noise bandwidths the thresholds refer to.

LINEWIDTH CONVENTION

The detection statistic is the flux density in a single channel after drift correction, so the quoted $\mathrm{EIRP}_{5\sigma}$ applies to any transmitter narrower than the native channel with the threshold flat in the assumed linewidth. Two sub-channel qualifications: the flatness is exact for a tone centred in a channel (a boundary-straddling tone divides its

TABLE 11
Survey vocabulary.

Term	Meaning in this paper
Census	every star within 40 pc with public ALMA archival coverage (168 entries; §3)
Bycatch	census stars observed by ALMA for unrelated science goals
Target/band dataset	one star in one ALMA band; the unit of the 208-dataset plan (104 complete)
Spectral window	one correlator spw of one execution block, after deduplication; the unit of search and bookkeeping (417)
Execution block (EB)	one contiguous ALMA observing session (100 searched; Appendix A)
MOUS	member observation unit set: the ALMA data hierarchy level at which pipeline products are delivered (one calibration recipe per MOUS)
QA2	the ALMA observatory pipeline's second-stage quality-assurance calibration and scoring, whose products this survey consumes
`obscore`	the IVOA table protocol of the ALMA Science Archive metadata service from which coverage and EB metadata are queried
Native channel	the correlator channel width of the window, 15.3 kHz–31.25 MHz here
Threshold crossing ("hit")	a $\geq 5\sigma$ spectral feature at the star's position in a searched window
Candidate	a threshold crossing promoted by the joint threshold-and-localisation criterion for manual vetting (three)
Control ring	512 off-source test positions per window, used for empirical false-alarm calibration
Star-beats-ring	a window whose star statistic exceeds all 512 control statistics (5/417)
Drift trial	one trial Doppler drift rate of the stack grid; 2–1944 per window
Nominal threshold	noise-derived 5σ detection threshold, *not* a calibrated completeness limit (boxed rule, §4)
$\mathrm{EIRP}_{5\sigma}$	equivalent isotropic radiated power at the nominal threshold; $4\pi d^2 S_{\min}\Delta\nu_{\mathrm{ch}}$
t_{on} / T_{span}	summed on-source integration / elapsed block span per star (Table 21)
Molecular-line exclusion mask	the $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame region around catalogued transitions where crossings are excluded (§4.2)
Closure phase	interferometric phase triplet test discriminating point-source structure (Appendix F)
Stellar frame	topocentric $\rightarrow$ barycentric $\rightarrow$ systemic-RV frame chain (§4.2)
Funnel	the candidate-narrowing cascade from channels searched to vetted candidates (Table 4)

power between two channels, rising toward a factor of two in the worst case; the measured dependence is given in Appendix E); and the transition between adjacent channels follows the correlator's channel passband shape (including any Hanning weighting or spectral averaging), which varies with correlator generation and is not retained in the frozen products, so the flat-in-linewidth statement is a channel-centred idealisation, accurate at the tens-of-per-cent level. A robustness test with injected tones at randomised sub-channel positions is committed with the validation items of §6. For the deepest window (UV Ceti Band 3) the quoted limit is 1.6×10^{13} W at a 15.625-MHz channel and the minimum detectable spectral power is $1.0 \times 10^{6}\,\mathrm{W\,Hz^{-1}}$: a 1-Hz carrier there needs the full 1.6×10^{13} W. Had the same observation been channelised at 1 Hz from the start, the threshold would fall as $P_{\min} \propto \sqrt{\Delta\nu}$ to 4×10^{9} W; archival data cannot be re-channelised below native resolution, so the ratio $\sqrt{\Delta\nu_{\mathrm{ch}}/\Delta\nu'}$ (∼4000 for that window) is the unavoidable price of the archival channelisation .

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3. The per-channel rms of the searched window is 0.244 mJy, so $S_{\min} = 5 \times 0.244 = 1.22$ mJy in a single 15.625-MHz native channel. At 2.67 pc this corresponds to $1.22 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$, and $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\min}\Delta\nu = 4\pi(2.67\,\mathrm{pc})^2 \times 1.22 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}} \times 15.625\,\mathrm{MHz} = 1.6 \times 10^{13}$ W. Any carrier narrower than 15.625 MHz must deliver that *total* power to be flagged in this window; a transmitter wider than the channel needs $S_{\min}$ per channel.

INTRA-INTEGRATION SMEARING

The frequency excursion of a signal at the maximum searched drift rate across one integration, $\dot{\nu}_{\max} t_{\mathrm{int}}$, is evaluated for every window: across the 417 windows it is at most 1.10 channels (median 0.001, 90th percentile 0.06). A linearly drifting signal smeared by Δ channels within an integration retains a fraction $\eta_{\mathrm{smear}} = \mathrm{sinc}(\pi\Delta/2)$ of its peak-channel amplitude, so the effective threshold for a worst-drift transmitter is $\mathrm{EIRP}_{5\sigma}/\eta_{\mathrm{smear}}$; the three affected windows of 417 are those of Table 12 and the median window is essentially unsmeared. A signal drifting midway between two adjacent trial rates accumulates at most 0.03 channels of residual drift across the grid as realised, so no significant sensitivity is lost to signals falling between trial rates.

PRIOR LOW-FREQUENCY TECHNOSIGNATURE SURVEYS

Table 13 summarises the surveys cited in §2. Content is limited to what the cited abstracts state; a dash marks quantities the cited work does not quote. Band, backend, and threshold conventions differ across rows by construction, so the table is context, not a sensitivity ranking (Figure 1).

TABLE 12
THE ONLY WINDOWS OF THIS RELEASE WITH NON-NEGLIGIBLE INTRA-INTEGRATION SMEARING AT THE MAXIMUM SEARCHED DRIFT RATE ($\eta_{\rm smear} < 0.99$; ALL 222 OTHER WINDOWS HAVE $\eta_{\rm smear} \geq 0.99$ AND THEIR NOMINAL AND EFFECTIVE THRESHOLDS AGREE TO BETTER THAN 1%). Δ IS THE FREQUENCY EXCURSION IN CHANNELS ACROSS ONE INTEGRATION, $t_{\rm int}$ THE PER-INTEGRATION ON-SOURCE TIME, AND $\mathrm{EIRP}_{5\sigma,\rm eff}$ THE SMEARED EFFECTIVE THRESHOLD $\mathrm{EIRP}_{5\sigma}/\eta_{\rm smear}$.

Window	$\Delta\nu_{\rm ch}$ (kHz)	$t_{\rm int}$ (s)	Δ (ch)	$\eta_{\rm smear}$	$\mathrm{EIRP}_{5\sigma,\rm eff}$ (10^{13} W)
β Pic B6	15.3	6.05	1.096	0.574	4.08
η Crv B8	61.0	6.05	0.585	0.865	365.8
η Crv B7	61.0	6.05	0.411	0.932	4.68

TABLE 13
SELECTED PRIOR TECHNOSIGNATURE PROGRAMMES BELOW 10 GHz AND THE SOLE PRIOR ALMA SEARCH, AS CITED IN THE TEXT.

Programme / study	Facility	Scale	Frequency	Note as stated by the cited work
Project Ozma (Drake 1960)	—	2 stars	1.42 GHz	τ Cet and ϵ Eri
Project Phoenix (Backus & Project Phoenix Team 2002)	multi-site	—	—	pioneered multi-site RFI verification
Allen Telescope Array (Tarter 2001)	ATA	—	—	first purpose-built commensal SETI instrument
ATA TRAPPIST-1 (Tusay et al. 2024)	ATA	—	—	—
ATA ISO survey (Sheikh et al. 2025b)	ATA	—	—	—
BL first data release (Enriquez et al. 2017)	GBT	692 stars	1.1–1.9 GHz	EIRP $> 10^{13}$ W
BL 1327-star search (Price et al. 2020)	—	1327 stars	1.10–3.45 GHz	—
BL Earth Transit Zone (Gajjar et al. 2019)	GBT	—	3.95–8.00 GHz	—
Margot et al. (Margot et al. 2023)	GBT	11,680 stars	1.15–1.73 GHz	—
ML vetting (Ma et al. 2023)	GBT	820 stars	—	β-CVAE cuts human-vetting load by $\sim$100$\times$
turboSETI correction (Choza et al. 2024)	—	—	—	S/N mischaracterisation; retrospective corrections
FAST commensal ISOs (Zhang et al. 2020; Li et al. 2026)	FAST	—	—	interstellar-object targets
Occultation timing (Tusay et al. 2025)	—	—	—	eclipsing/occultation strategy
COSMIC (Tremblay et al. 2024)	VLA	—	—	commensal system
MeerKAT dragnet (Czech et al. 2021)	MeerKAT	—	—	—
MWA back-ends (Morrison et al. 2023)	MWA	—	—	—
ALMA stellar bycatch (Mason et al. 2024)	ALMA B3	28 stars	90.642/93.151 GHz	first ALMA technosignature search; $\mathrm{EIRP}_{\rm min} = 6.9 \times 10^{17}$ W (nearest star)

TABLE 14
SCOPE OF THE ANALYSES IN THIS RELEASE: WHAT EACH WAS APPLIED TO, ITS VALIDATION STATUS, AND WHETHER IT IS USED IN THE HEADLINE LIMITS.

Analysis	Applied to	Validation status	In headline limits?
Primary narrowband search	417 windows / 85 stars (79 systems)	Injection-calibrated for the searched class (1200 trials; Appendix E); false-alarm rate calibrated empirically (§5.4)	Yes
Closure-phase vetting	3 candidates	Simulation-validated; sensitivity-limited on sky (Appendix F)	Disposition only
Injection-recovery validation	6 configurations / 3 stars	This release; scope limits stated (Appendix E)	Completeness only
Chirp/periodicity screen	122 retained spectra	Demonstration-grade (Appendix G)	No
Frequency-occupancy check	366 overlapping star pairs	Executed (Appendix H)	Disposition only

TABLE 15
SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS. CENSUS: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ AND $1/\varpi \leq 40\,\mathrm{pc}$ (17566 OBJECTS; 8594, 49 PER CENT, CARRY A `teff_gspphot` ESTIMATE). SAMPLE: THE 85 STARS WITH AT LEAST ONE SEARCHED WINDOW IN THIS RELEASE. SAMPLE DISTANCES ARE THOSE USED BY THE SEARCH ITSELF AND ARE COMPLETE; TEMPERATURES EXIST FOR 51 OF THE 85 STARS (ALL 85 MATCHED TO THE 168-ENTRY ALMA-COVERED CENSUS, 56 BY EXACT NAME; 51 OF THOSE MATCHES CARRY A TEMPERATURE), SO THE TEMPERATURE PANEL IS NORMALISED WITHIN EACH COLUMN OVER CLASSIFIED OBJECTS ONLY AND T_{eff} PROXIES SPECTRAL CLASS. "RATIO" IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION: 1 MEANS THE SAMPLE TRACKS THE REFERENCE POPULATION, >1 OVER-REPRESENTATION.

Property	Census	Searched	Ratio
Distance (all 85 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (14.1%)	51.7
5–10 pc	240 (1.4%)	13 (15.3%)	11.2
10–20 pc	2069 (11.8%)	18 (21.2%)	1.8
20–30 pc	5391 (30.7%)	18 (21.2%)	0.7
30–40 pc	9818 (55.9%)	24 (28.2%)	0.5
T_{eff} class[a] (51 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (2.0%)	7.0
F (6000–7500 K)	402 (4.7%)	9 (17.6%)	3.8
G (5300–6000 K)	860 (10.0%)	15 (29.4%)	2.9
K (3900–5300 K)	1400 (16.3%)	5 (9.8%)	0.6
M (< 3900 K)	5908 (68.7%)	21 (41.2%)	0.6
Other axes			
Known planet hosts[b]	20 (11.9%)	18 (21.2%)	1.8
Searched stars / systems	–	85 / 79	–

[a]Bins are temperature bins, not MK types: A $\geq$ 7500, F 6000–7500, G 5300–6000, K 3900–5300, M $<$ 3900 K. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios correspondingly upper limits. White dwarfs are not separable here and fall wherever their (unreliable) photometric temperature places them.

[b]The Gaia reference census carries no planet information, so the census column here is the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 85 searched stars include 18 hosts of 41 known planets.

SUPPLEMENTARY MATERIAL (ONLINE ONLY)

The appendices below are online-only supplementary material, not part of the printed article; they are carried in the preprint source so that every number in the main text is reproducible. Each, with each relocated figure and table, is cross-referenced where the main text uses its summary.

INJECTION-RECOVERY SENSITIVITY VALIDATION

The limits quoted here assume a 5σ threshold behaves as an idealised Gaussian-noise calculation predicts. We test that by injecting synthetic narrowband signals of known amplitude into real calibrated visibility data, as far upstream of the statistic as the retained products allow and through the extraction code's own mechanism, and measuring what the unmodified pipeline recovers. An early 24-tone Proxima Cen pilot (zero-drift only) put the 50 per cent recovery point near 6σ; the campaign below supersedes it and reproduces that point as threshold marginality.

The campaign injects into six window configurations spanning three bands and both resolution classes – three coarse Band 7 spws of AT Mic, one coarse Band 3 and one coarse Band 6 spw of AU Mic, and AU Mic's 488-kHz flagband window (49-trial drift grid) – placing a complex tone at the star position at amplitudes 4–10× the statistic's own noise, drift fractions f_{drift} = 0, 0.25, 0.5, 0.75, 1.0 of each window's searched ceiling, on- and half-grid rates, and channel-centred and boundary placements: 200 trials per configuration, 1200 in all, run through the unmodified search

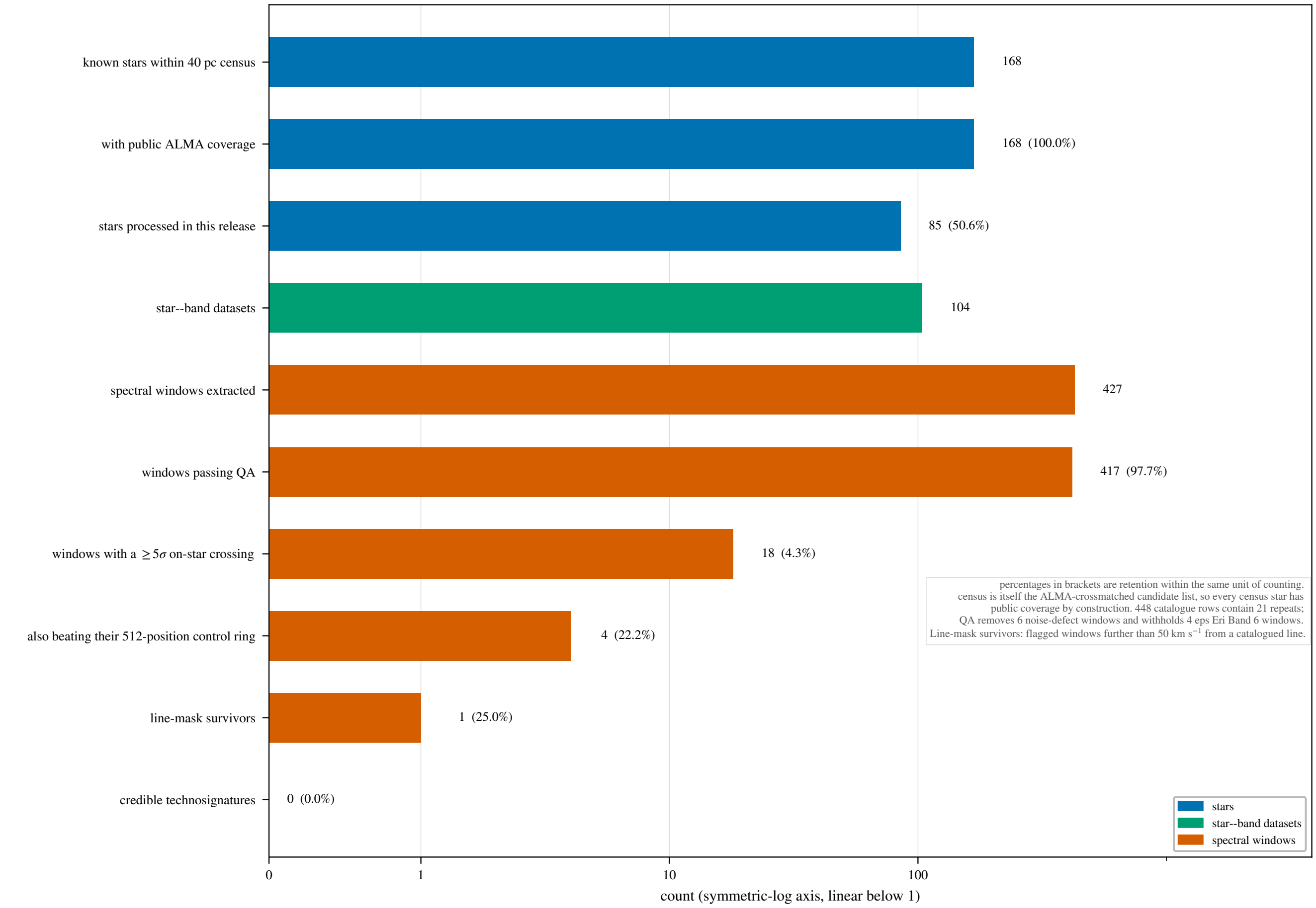

Fig. 9.— The survey's selection funnel, from archive availability to the zero-candidate outcome. Top: star selection. Bottom, the archive-availability accounting, which has been run on the ≤20 pc subset and quotes that subset's counts: public on-star execution blocks available (all bands; proper-motion-matched archive census of 2026 September 9), the star–execution-block pairs within scope, the window-level records with their correct unit (∼4 spectral-window rows per pair), and the search outcome. The searched release as a whole spans 100 execution blocks. The difference between availability and scope is the recurrence pool drawn on by the AU Mic second-epoch test (§5.4).

TABLE 16
Consolidated family-wise trials accounting for this release. "Absorbed by" names the step carrying each factor's multiplicity in the operative false-alarm model; the bookkeeping ladder is detailed in Appendix P.

Trials factor	Unit	Count
Channels per window (median; maximum)	window	118; 3770
Drift trials per window (median; maximum)	window	4; 1944
Channel × drift trials	release	3.97×10^7
Effectively independent channel cells[b]	release	8.5×10^4–1.1×10^5
Control positions / $N_{\rm eff}$	window	512 / ∼30–43
Windows (star-level tests)	release	417
Windows containing an on-star hit	release	18
Gaussian-expected chance crossings	release	11.4
Cross-target frequency pairs tested	release	946
Star-beats-ring rate $\hat{p}_{\rm sbr}$	window/release	64/417 (15.3%)
Execution blocks spanned	release	100
Principal false alarm: EB-block permutation	release	$P(F{\geq}3) = 6.1\%$ ($E[F]$=1.04)
Expected flags under the symmetric null ($N/513$)	snapshot	0.81
Wilson-rate bracket on $P(F{\geq}3)$	release	4–19%

[b]From the measured lag-one channel autocorrelation of the drift-tracked statistic, $\rho_{\rm lag1} = 0.989$–0.991.

with the recovery criterion frozen beforehand ($T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step; Fig. 11).

The first result is structural and defines the searched class. The pipeline's per-channel time-median subtraction, which removes each position's continuum, also absorbs any phase-steady carrier dwelling in one channel for more than roughly 40 per cent of the track. On the coarse windows this is total: within the drift ceiling every drift is sub-channel over the track, so *every* in-grid carrier is static in this sense, and the pipeline-identical variant recovers 0

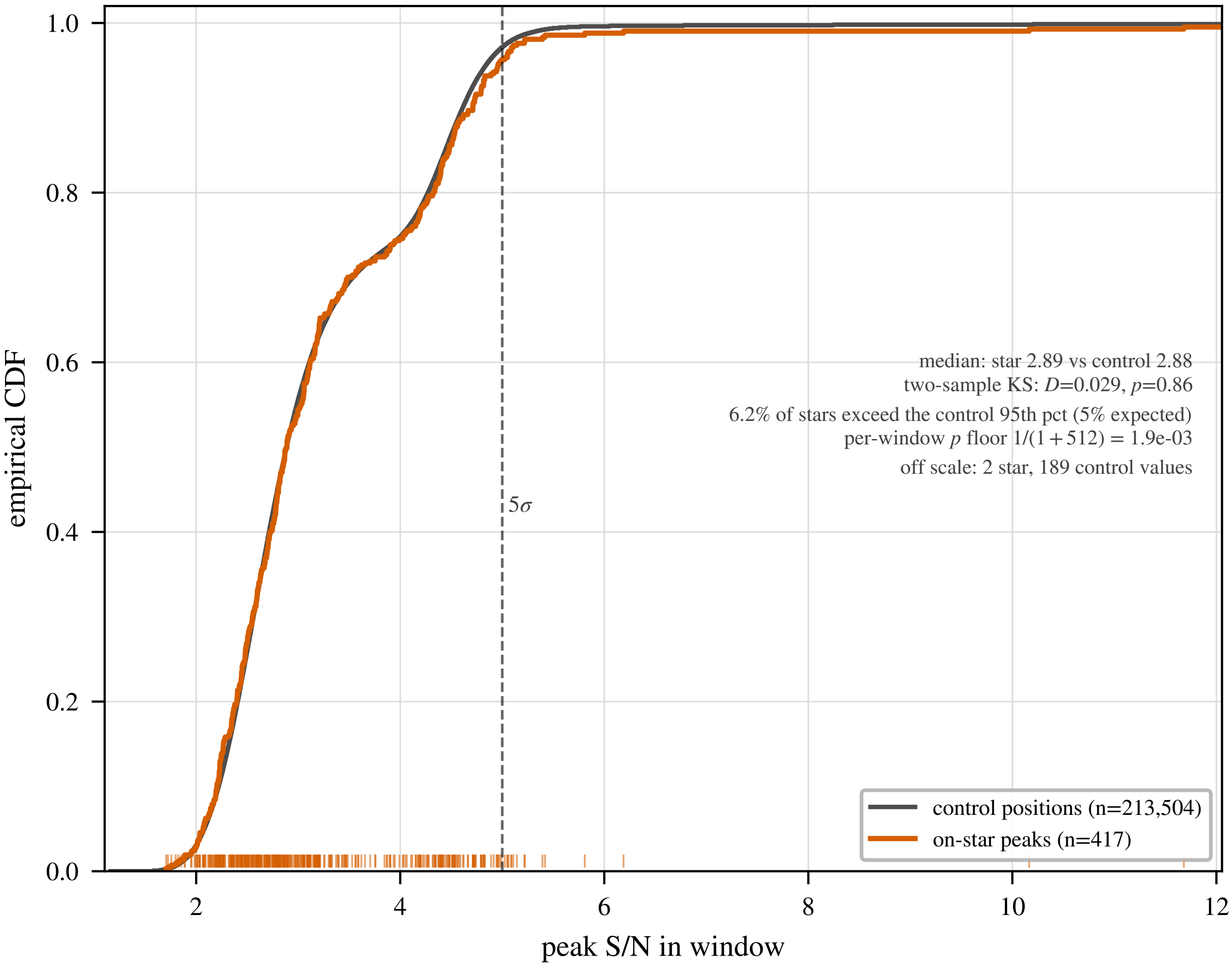

FIG. 10.— The symmetric star-versus-control null on the seven locally reprocessed windows (six validation windows plus the AU Mic candidate window). Each window's p is the add-one rank of the star statistic $T_\star$ against the 512 control statistics under identical treatment, and is Uniform(0,1) under the null. All seven lie at $p \geq 0.81$; with $n = 7$ the 95 per cent Kolmogorov band is wide, so the figure's content is the absence of star-side excess, not a precise distribution. The control ring's effective independence is $N_{\rm eff} \simeq 30$–43.

of 500 coarse-window injections at every amplitude from 4σ to 10σ. The coarse-window EIRP$_{5\sigma}$ limits therefore bound transmitters time-variable on track timescales – amplitude-, phase-, or duty-cycle-modulated below the $\sim$50 per cent dwell threshold – and say nothing about phase-steady carriers. The fine window measures the same boundary in drift space: $f_{\rm drift} \leq 0.25$ carriers (dwelling $\geq$40 per cent of the track per channel) are absorbed, while $f_{\rm drift} \geq 0.5$ carriers survive and constitute the searched class.

The second result is the completeness curve of the searched class. For drifting carriers on the fine window ($f_{\rm drift} \geq 0.5$, pooled over grid phase and sub-channel placement), the pipeline recovers 17, 42, 58, 75, and 83 per cent at 4, 5, 6, 8, and 10σ: roughly half at the nominal threshold, saturating by twice the threshold power. The search machinery alone (tone injected after the subtraction) is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones at every drift fraction; boundary-straddling placement costs a factor 0.6–0.8 in the statistic, and resolved channel-phase-wise, recovery at 5σ is 83 per cent for channel-centred tones and 0 per cent for boundary-straddling ones (100 and 67 per cent at 10σ), the 50-per-cent crossing falling near 1.4$\times$ threshold amplitude and full recovery at $\sim$1.6$\times$ threshold power. The survey completeness uses the placement-pooled curve above, an average over sub-channel phase and conservative for the boundary-straddling case (§6). For the fine window's searched class, EIRP$_{50}$ = 1.10 EIRP$_{5\sigma}$ (the interpolated half-recovery point, 5.5σ) and EIRP$_{90}$ > 2.0 EIRP$_{5\sigma}$ as a bound, the curve reaching only 83 per cent at the largest amplitude injected so the 90-per-cent point lies beyond the measured range. Coarse windows carry no EIRP$_{50}$ for the searched class by construction, and Table 17 collects the per-class reading.

The campaign's scope is configuration-level: six configurations on three stars, covering Bands 3, 6, and 7 directly. The Band 8 windows share the coarse-channel construction, over which the static-carrier result holds by the same mechanism, but their drifting-class completeness is transferred by that construction rather than measured (§6). Not spanned: other target environments, primary-beam-offset positions, integration durations, and edge-phased drift grids;

TABLE 17
Per-class reading of the quoted limits. $C_{5\sigma}$ is the measured recovery of the class at the nominal threshold.

Window class	Searched class	$C_{5\sigma}$	EIRP_{50}	Static-carrier recovery
Fine ($\leq$1 MHz)	drifting	0.42	1.10 $\mathrm{EIRP}_{5\sigma}$	0/20 (absorbed)
Coarse (15.625 MHz)	time-variable	not calibrated[a]	...	0/500 (by construction)

[a]Every in-grid drift on a coarse window is sub-channel over the track, so the injected (constant) tones were all phase-steady; recovery of the time-variable modulation defining the coarse searched class has not been measured. Static carriers are suppressed on both classes.

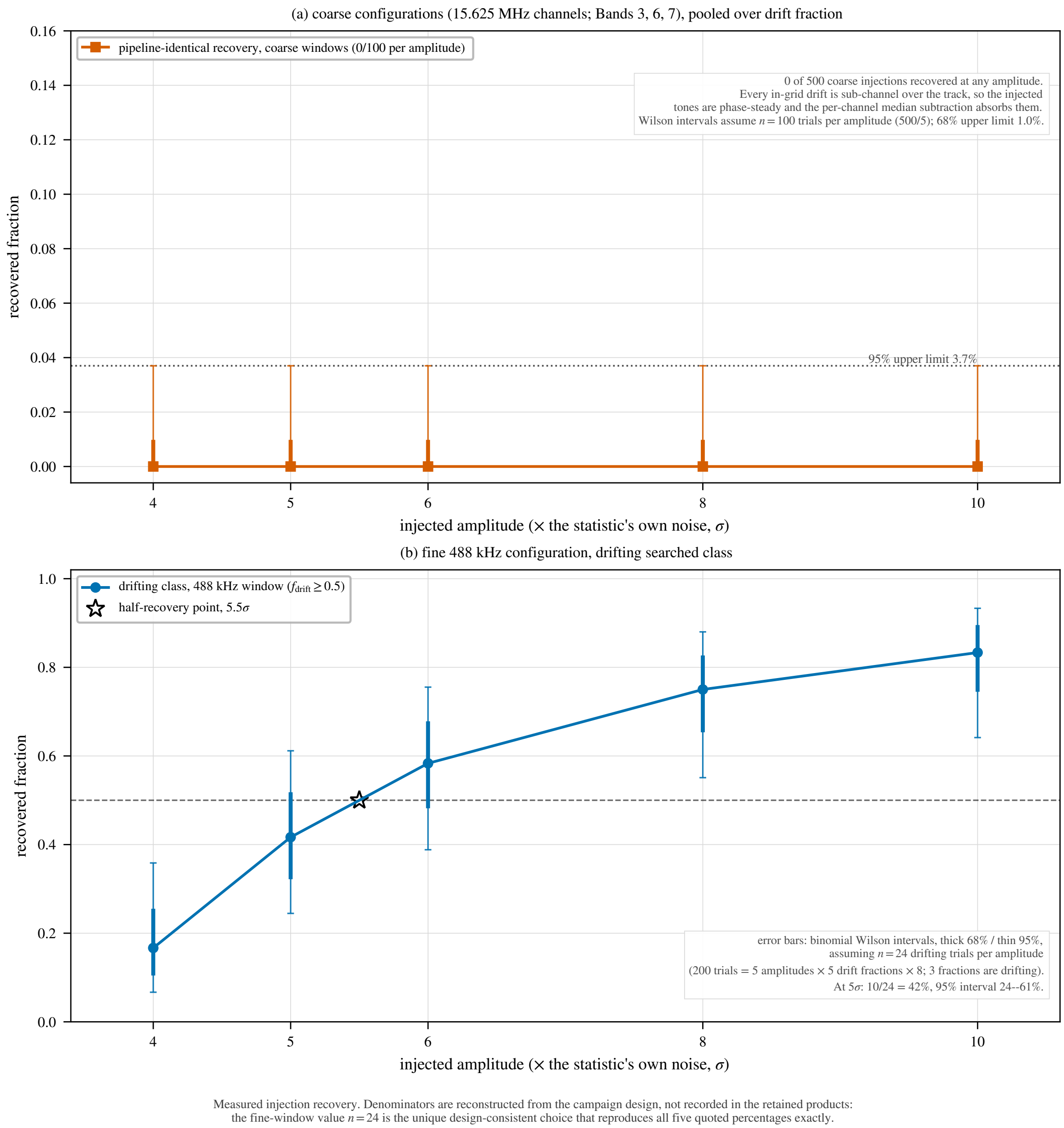

Fig. 11.— Measured injection recovery on real ALMA data (1200 trials, this appendix). Top: the five coarse-channel configurations (15.625 MHz channels; Bands 3, 6, 7), pooled over drift fraction and grid phase, where the pipeline-identical variant (solid) recovers nothing by construction and the machinery's detection fraction (dashed; $T \geq 5$ and peak channel within 3 of the injection) is the meaningful remaining quantity. Bottom: the 488-kHz-channel configuration, where the drift dimension is real (49-trial grid) and the pipeline-identical recovery $C(A, f_{\mathrm{drift}})$ rises with amplitude for drifting carriers ($f_{\mathrm{drift}} \geq 0.5$) while static and near-static ones ($f_{\mathrm{drift}} \leq 0.25$) are absorbed. Boundary-straddling tones can fail the frozen criterion's peak-drift clause at high drift while still being detected, the origin of the non-monotonic drift dependence near the ceiling.

campaigns on Band 8 data and on the β Pictoris windows are infeasible with the measurement sets retained locally for this release, and are declared as such rather than approximated.

CLOSURE-PHASE POINT-SOURCE VETTING: SUMMARY AND FULL SPECIFICATION

Threshold-crossing candidates face a further automated test that single-dish and beamformed surveys lack, since it needs multiple independent baselines. Its sensitivity limit at this survey's flux densities is stated in §4.4 and quantified below; the release's three candidates are all consistent with a point source (z = 0.16 for AU Mic; §5.4).

The closure phase is the phase sum $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ around a closed triangle of baselines i-j-k. It is exactly zero for an unresolved point source at any sky position, because such a source's visibility phase is a linear function of (u, v) that cancels around any closed loop, and it is immune to per-station calibration errors. Being independent of source position, it asks only whether the signal behaves like a coherent point source at all: a coherent far-sidelobe interferer (satellite, aircraft) would itself produce near-zero closure phase and pass.

Consistency with zero is assessed by a z-score, the mean drift-tracked closure phase over its standard error across the closed triangles, with three properties handled explicitly. First, triangles of one observation are not independent, since any two sharing a baseline share its thermal noise, so the independent closure quantities number fewer than the triangles (of order 10^4 for a 40-antenna configuration) and the aggregate score must not treat triangles as independent trials. Second, closure phases wrap at $\pm\pi$, so the statistic is circular (unit-phasor averaging). Third, its null distribution is calibrated empirically against the observation's own integrations rather than assumed Gaussian, which also absorbs the shared-baseline covariance.

The calculation was validated against synthetic visibility data before any ALMA application: zero closure phase for a noiseless simulated point source, consistency with zero for a noisy one, clearly inconsistent scatter for a simulated RFI-like non-point-source signal. The positive control drove continuum detections through the closure machinery at their detection frequencies, on four fields of the local reprocessing set (AT Mic Band 7, the known ∼2-arcsec binary, as the resolving control; AU Mic Band 6 and Band 3 continuum; and the AU Mic Band 6 candidate frequency 230.8252 GHz), taking triangle closure phases from time- and channel-averaged unflagged cross visibilities and propagating $\sigma_{\rm CP}$ from per-baseline integration-to-integration scatter. The result is a sensitivity verdict: per-baseline S/N is 0.5–0.7 at the median and never exceeds 5.1 (0.02 per cent of triangles have all three baselines above S/N = 5), so every field's closure phases are indistinguishable from pure noise, including the resolved binary, whose ∼0.4 mJy total flux sits below the per-baseline noise of these short integrations. The test triggers only on threshold crossing, at negligible cost.

CHIRPED-DRIFT AND PERIODICITY SCREEN: SUMMARY AND FULL SPECIFICATION

The primary search stacks at constant linear drift rate, so it is by design blind to a transmitter whose drift rate changes during the observation and to pulsed or periodically-modulated carriers. We implemented a chirped (quadratic-drift) extension of the same drift-corrected stacking search, on a coarser grid of 21 linear-drift × 5 chirp-rate trials, and a per-channel time-domain periodicity search: an FFT along each channel's time axis, per position, with the peak power compared against a robust per-channel noise estimate. Both vet the star's dynamic spectrum against 8 retained control positions, as the primary search vets the star against its control ring; those 8 are a strict subset of that window's 512-position ring, not an independent ensemble (§4), and are retained for every target processed to date, so analyses like these need re-extract nothing.

Applied to the 122 retained spectra, with a 9-member ensemble and coarser grids, this is a demonstration rather than survey-grade: its role is to establish that the extension runs and behaves correctly under the null. It does. The exceedance rates (10.7% chirp, 13.1% periodicity, against the ∼11% chance rate of a star being the maximum of a 9-member ensemble) are as expected, no instance is treated as a candidate, and no limit derives from it. Extending both to the full 512-position control ring, at negligible cost since the spectra are already retained, is what promotes the screen to survey grade, and is deferred to the completed survey.

CROSS-TARGET FREQUENCY-OCCUPANCY CHECK: SUMMARY, KERNEL, COORDINATES, AND SCOPE

This single-pointing archival survey has no ON/OFF cadence for rejecting radio-frequency interference; its analogous resource is the many unrelated stars sharing similar correlator setups, since a narrowband feature recurring at the same observed *sky frequency* across unrelated targets is almost certainly RFI or instrumental. We cross-match threshold crossings towards different targets falling within $k = 3$ native channels of the coarser width (kernel below), so the null rejects sky-frequency recurrence only. Across the 417 windows and 107 tabulated crossings, no pair falls within the kernel, or within 0.5 MHz. The kernel (∼46 kHz) is narrower than the β Pictoris line offsets of 10.6 and 21.9 MHz, so those candidates were dispositioned by line-catalogue inspection, not by this check.

Matching kernel. The check cross-matches, across every target/spw result in this release, both the best-fit-channel frequency (whether or not it formally crossed the detection threshold) and the list of formal $\geq 5\sigma$ crossings, pairing two crossings at frequencies ν_1, ν_2 towards different targets when $|\nu_1 - \nu_2| < \max(k\,\Delta\nu_1,\ k\,\Delta\nu_2)$, so the kernel scales with the coarser of the two native channel widths rather than being a fixed interval. We adopt $k = 3$ native channels; the 5 MHz tolerance of the original formulation corresponds to k between ∼0.2 and ∼330 across this release's native channel widths (15.3 kHz to 31.25 MHz).

The three frequency coordinates. Three coordinates are at play: the *sky frequency* of a hit as reconstructed from the archive spectral-window metadata in the observation's recorded frame; the *topocentric frequency*, which differs from a barycentric convention by up to the $\pm 30\,\mathrm{km\,s^{-1}}$ barycentric term (∼23 MHz at 230 GHz, far wider than the matching kernel); and the *correlator channel index* of the native spectral window. The sky-frequency cross-match probes the first, the channel-index cross-match the third. A frame-convention inconsistency between two observations, or an artefact fixed in the second, would each defeat sky-frequency pairing, which is why the frame-convention disclosure queued

TABLE 18
Target pairs with the deepest searched-frequency overlap, from the merged per-window frequency ranges of this release. A shared tuning is the circumstance in which a recurring sky-frequency feature would be caught.

Target pair	Overlap (GHz)
γ Lep – γ Vir	14.25
γ Vir – TRAPPIST-1	10.78
γ Lep – TRAPPIST-1	10.77
HD 10647 – HR 1010	10.13
AU Mic – β Pic	8.18
GL 3379 – HD 285968	7.32
GJ 849 – LHS 1140	7.32
TRAPPIST-1 – Wolf 359	7.31
Proxima Cen – Wolf 359	7.31
Proxima Cen – TRAPPIST-1	7.31

with the symmetric-null recomputation (§5.4) feeds this check too. Artefacts fixed in the correlator frame – correlator birdies, local-oscillator spurs, quantization spurs, bandpass-edge ripple – recur at the same intermediate frequency or channel number rather than the same sky frequency; a channel-index cross-match (same channel number of an identical correlator setup) is possible with the retained products and also returns zero pairs, while a full intermediate-frequency reconstruction, which would pair artefacts across different tunings, is not recoverable from the retained metadata and remains with the completed survey.

Exposure and exclusions. The power of the null is the count of target pairs sharing frequency coverage, from the per-window frequency ranges of the narrowband-covered stars, on the $\leq$20 pc subset over which this check has been executed: 366 of the 946 star pairs (39%) overlap in searched frequency by at least 0.01 GHz – 359 (38%) by at least 0.5 GHz and 345 (36%) by at least 1 GHz – and 109 overlapping pairs involve one of the five stars carrying formal crossings, so every crossing frequency was tested against genuinely overlapping independent targets; Table 18 lists the deepest overlaps. Two execution blocks are shared between catalogue entries, the UV Ceti binary's (one field, carried as the G 272-61A/B entries) and the AT Mic pair's (AT Mic B and AT Mic AB; §3), and their intra-pair "coincidences" are identities by construction, excluded from the cross-target test.

One bookkeeping caveat bounds the null on that subset. The per-window hit lists of the interim retained product are capped at 50 frequencies: the true count over the subset is 171 crossings against 107 tabulated, the difference being one window, the β Pic Band 6 ultra-fine window, with 114 formal crossings of which the 50 strongest are tabulated. Those 50 span 0.748 MHz; the four crossings towards HD 285968, the only other star with crossings in that frequency region, lie 13.7 MHz from that extent but inside the window's full 54-MHz searched span, so the 64 untabulated crossings, whose individual frequencies the retained products do not carry, could not be tested for coincidence. The null is therefore demonstrated for the tabulated set and unresolved for the remainder – a limitation of the retained product, not of the criterion, which the public release removes by shipping uncapped hit lists (Data Availability).

PER-TARGET RESULTS

These results are data rather than argument, and at this scope a typeset table would run to many pages, so they are released alongside the manuscript source as `per_target_results_v3.28.csv`, and summarised here.

The released file has one row per extracted spectral window (448 at the snapshot epoch, each with its quality-control status), carrying star name and heliocentric distance, ALMA band and execution block, searched frequency range and native channel width, the nominal $\mathrm{EIRP}_{5\sigma}$ threshold, the line-excluded continuum rms, the stellar and control-ring peak signal-to-noise ratios, the count of controls reaching or exceeding the star, the empirical p-value of §5.4, the drift ceiling, on-source time, and the nearest catalogued transition with its velocity offset. Every quantity in this paper is reproducible from it.

$\mathrm{EIRP}_{5\sigma}$ is the nominal threshold, the minimum EIRP that would have produced a 5σ detection in that window, $\mathrm{EIRP}_{5\sigma} = 4\pi d^2(5\sigma_{\mathrm{comb}})\Delta\nu$, on the same threshold and primary-beam correction as the continuum column, so the two limits in each row are comparable. The per-channel flux-density threshold follows as $S_{\min} = \mathrm{EIRP}_{5\sigma}/(4\pi d^2\,\Delta\nu_{\mathrm{ch}})$, the narrowband per-channel noise and *not* the tabulated rms, which is the same target's line-excluded full-bandwidth continuum noise.

Bands 6 and 7 supply 83 per cent of the windows, an imprint of the archive rather than a design choice. Band 8 windows are an order of magnitude less sensitive in EIRP at fixed distance, and Bands 4 and 5 have a single star each, so neither carries weight in the aggregate limits.

THE VELOCITY-SPACE LINE MASK: JUSTIFICATION AND RETROSPECTIVE APPLICATION

The one-channel line check the searches ran was inadequate for the β Pictoris candidates (§5.4): a native channel is a fraction of a $\mathrm{km\,s^{-1}}$ wide, while ordinary circumstellar kinematics displace real emission by many channels. Its replacement, a velocity-space mask, takes its tolerance from the kinematic budget of real circumstellar gas: Keplerian motion where these molecules survive ($\gtrsim$10 au) is $\lesssim$13 $\mathrm{km\,s^{-1}}$, cold-gas intrinsic linewidths are $\lesssim$1 $\mathrm{km\,s^{-1}}$, the eight masked species are cold dense-gas tracers (no bulk-wind term for the dwarfs dominating the sample), and the uncorrected barycentric term is bounded by $\pm$30 $\mathrm{km\,s^{-1}}$. Empirically it is conservative: the largest attributed circumstellar component (β Pic, -2.7 to -3.6 $\mathrm{km\,s^{-1}}$) sits at 7 per cent of it, and *zero* candidate frequencies fall between

TABLE 19

PER-BAND SUMMARY OF THE SEARCHED WINDOWS AT THE SNAPSHOT EPOCH. n IS THE WINDOW COUNT, "STARS" THE DISTINCT STARS CONTRIBUTING, AND THE EIRP COLUMNS THE MEDIAN AND BEST NOMINAL 5σ THRESHOLDS. "FINE" COUNTS WINDOWS CHANNELISED BELOW 1 MHz, THE REGIME IN WHICH THE INJECTION CAMPAIGN RECOVERS THE SEARCHED CLASS.

Band	n	Stars	Median EIRP (W)	Best EIRP (W)	Fine
3	44	9	9.8×10^{13}	1.6×10^{13}	4
4	5	1	2.5×10^{13}	2.3×10^{13}	0
5	5	1	4.0×10^{13}	3.4×10^{13}	0
6	217	53	1.3×10^{15}	2.8×10^{12}	58
7	156	38	2.8×10^{15}	2.0×10^{12}	50
8	12	3	6.1×10^{16}	3.2×10^{15}	3

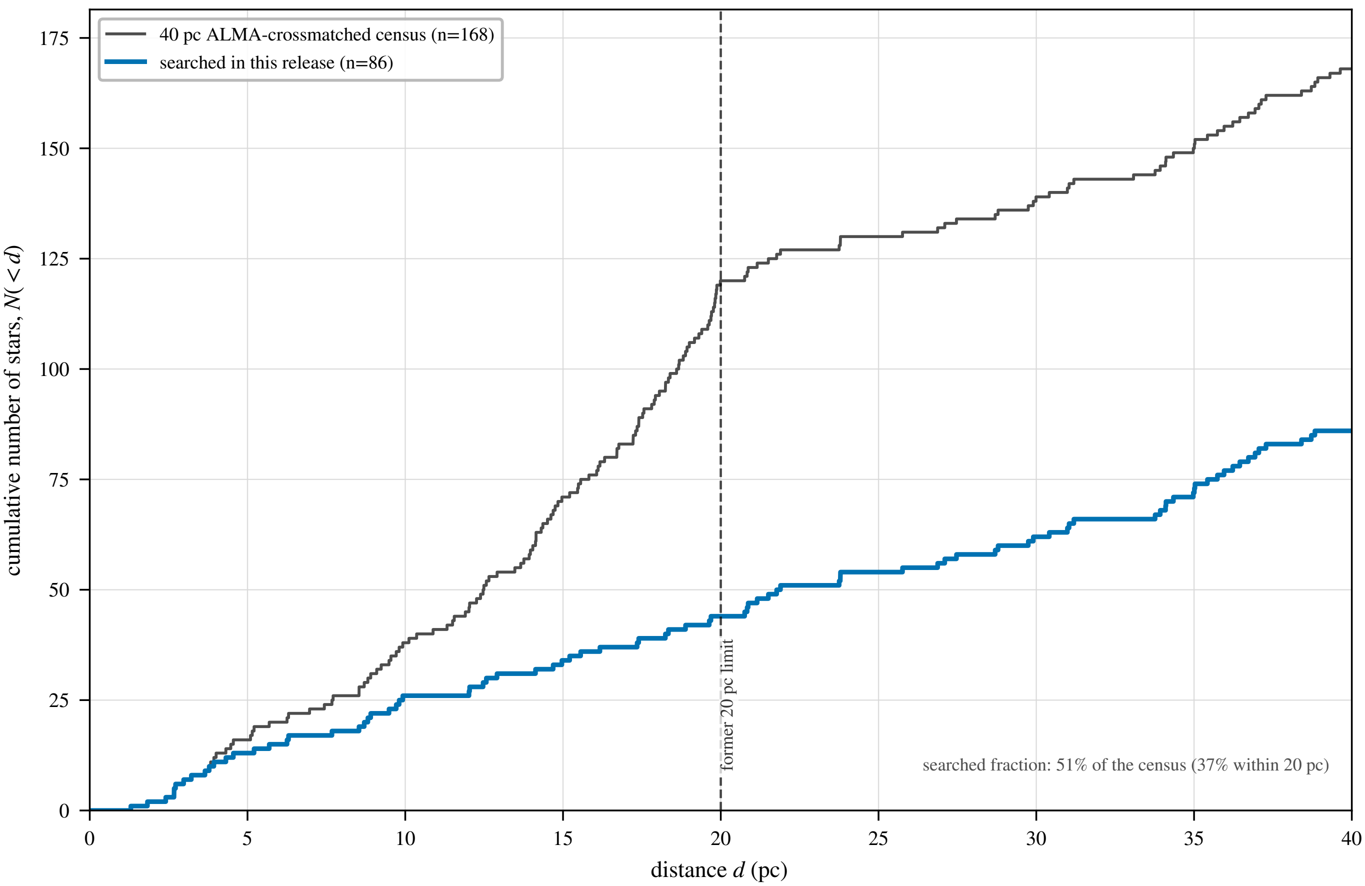

FIG. 12.— The ALMA-covered subset versus the stellar population within 20 pc: cumulative $N(< d)$. Grey: the reference Gaia DR3 catalogue (2357 stars); blue: this paper's census sample, only ∼5 per cent of the total. The curves are not expected to track one another (§3).

13 and $50\,\mathrm{km\,s^{-1}}$ from their nearest transition, so a $\pm 13\,\mathrm{km\,s^{-1}}$ Keplerian-only mask would have suppressed the same candidates and released the same five: the wider tolerance costs no candidate yield and buys robustness against radial-velocity catalogue error.

Applied retrospectively, the mask rejects exactly the two β Pictoris candidates and changes no other disposition. Of the remaining six crossing windows, two fall inside it (the β Pic coarse window without a candidate at $-3.6\,\mathrm{km\,s^{-1}}$, and HD 285968 at $+8.7\,\mathrm{km\,s^{-1}}$, already rejected by its own control ring, maximum 10.9 against a star peak of 6.5); the rest lie far outside any circumstellar tolerance (AU Mic +379, LHS 1140 $+115\,\mathrm{km\,s^{-1}}$). Both search products are reported: of the 18 windows containing an on-star hit, four also beat their control ring, and the mask accounts for three of those four, leaving the single unattributed crossing dispositioned by its spatial behaviour. One counter-argument deserves recording: the transitions that make astrophysical false positives cheap also make them natural "Schelling points" for a deliberate communicator (Cocconi & Morrison 1959). The released products therefore maintain a known-line-coincident list – every on-source 5σ crossing inside the masked intervals, with its offset from each nearby transition – so a future search with independent discrimination can revisit exactly the channels set aside.

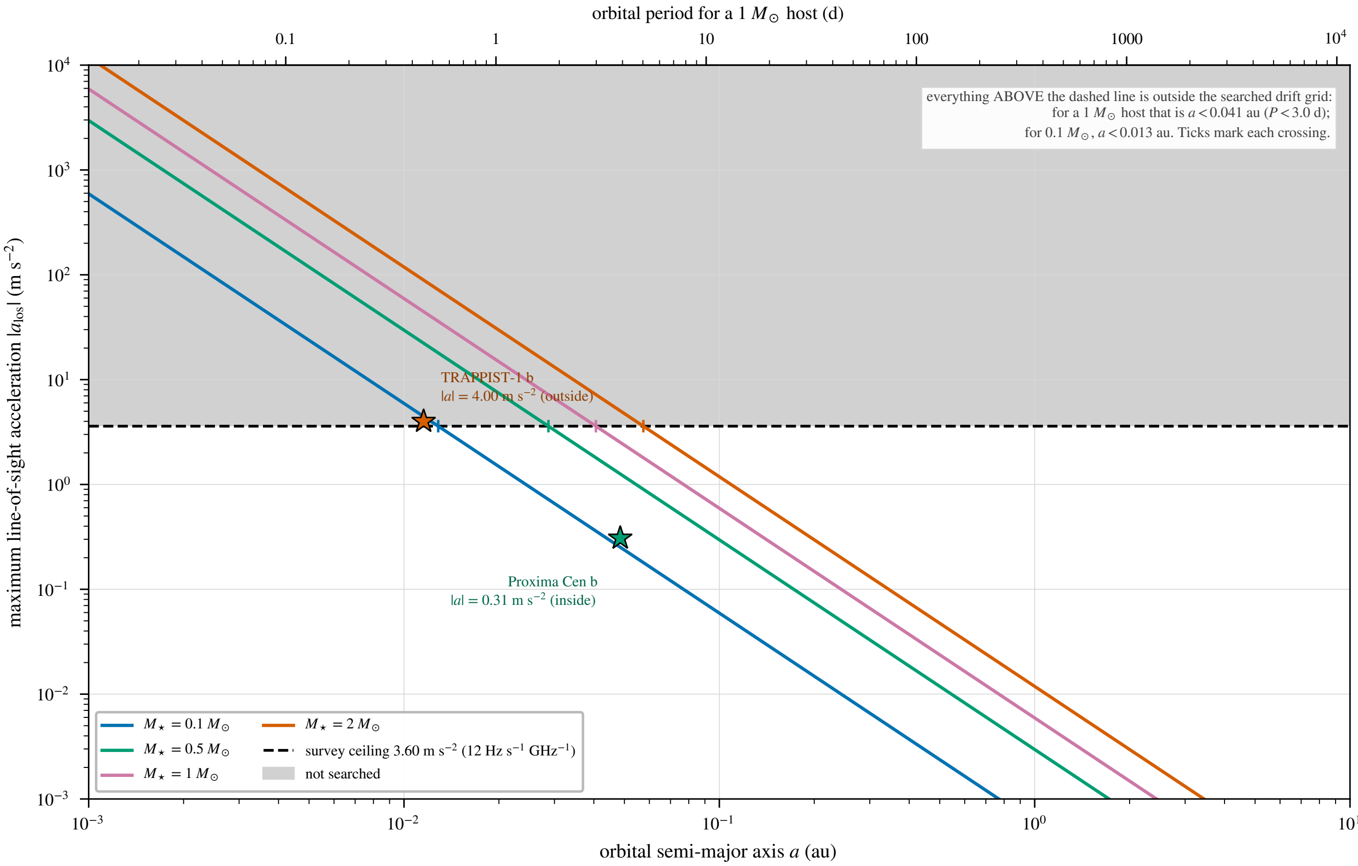

Fig. 13.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency), against orbital period, for host-star masses 0.09–1.8 $M_\odot$ (lines) and for the 37 known planets of the 15 planet-hosting targets in this release (points). The shaded band is the survey's drift-search ceiling (12–13 Hz s^{-1} GHz^{-1}; §5.4), above which a transmitter is outside the searched grid. Only TRAPPIST-1 b ($P = 1.51$ d) reaches it (§6).

TABLE 20

Per-band characterisation of the 417 searched windows. "Sys." counts unique stellar systems searched in the band, so a system observed in two bands enters both rows (79 systems, 85 stars in total); "EBs" the contributing execution blocks, attributed by the provenance of the de-duplicated windows; "Win." the searched windows; $\Delta\nu_{\rm ch}$ the native channel width in MHz (median, range in brackets); "Union" the unique sky frequency searched in the band after merging overlapping windows; EIRP$_{5\sigma}$ the median nominal threshold across the band's windows; $\dot\nu_{\rm ceil}$ the range of drift-rate ceilings in Hz s^{-1} per GHz of carrier frequency. The "All" row is computed over the whole sample, not summed down the columns: EBs and systems recur across bands and the frequency unions are disjoint by construction.

Band	Sys.	EBs	Win.	$\Delta\nu_{\rm ch}$ (MHz)	Union (GHz)	Med. EIRP$_{5\sigma}$ (W)	$\dot\nu_{\rm ceil}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	52	52	206	15.63 (0.015–31.25)	28.7	1.4×10^{15}	12.0–13.3
7	33	34	151	15.63 (0.061–15.63)	22.6	2.9×10^{15}	12.0–13.3
8	3	3	12	15.63 (0.061–15.63)	6.6	6.1×10^{16}	12.0
All	79	100	417	15.63 (0.015–31.25)	92.4	1.3×10^{15}	12.0–13.3

AU MIC: THE THREE TESTS IN FULL

Localisation. The first-epoch execution block was reprocessed from the raw science data model with the observatory pipeline's own calibration recipe, and the fully symmetric star-versus-control statistic of §5.4 run on the candidate window: $T_\star = 4.93$ against a control maximum of 7.70 ($p = 0.81$), so the star does not beat its control ring, under a reprocessing retaining 310 integrations against the release pipeline's 249 on a finer drift grid. The frozen products indicated the same asymmetrically: the source-region peak exceeds the 512-control maximum by only 0.12, less than the window-to-window scatter of that statistic, and the candidate fails the single-position variant too (5.22 against 5.85). This window therefore joins the six validation windows with the same outcome, no localised excess at the stellar position.

Recurrence. The archive holds exactly one further public Band 6 block covering 230.83 GHz, the same programme executed 2014 June 5, searched with treatment identical to the release: $T_\star = 4.77$ against a control maximum of 6.78

TABLE 21
Per-star observing duty cycle, ordered by distance, for the subset over which the epoch accounting has been compiled. The two spatially unresolved binaries (UV Ceti = G 272-61A/B; AT Mic = the B and AB entries) share every observation and appear as separate catalogue rows but are single trials, giving 79 independent systems. $N_{\rm win}$: searched spectral-window measurements in this release (417 in total, after deduplication); $N_{\rm EB}$: distinct execution blocks searched; $t_{\rm on}$: summed on-source integration in minutes, taking each execution block's median across its windows; $T_{\rm span}$: summed elapsed span of the execution blocks in hours, including calibration and overheads and so exceeding $t_{\rm on}$. These were reconstructed from the ALMA Science Archive (queried 2026 September 8) rather than retained by the frozen extraction products, and audited against the archive's independent exposure accounting (online-only Appendix Q), whose two exception groups (AT Mic and CD−38 10980) are carried at their archive-consistent exposures. Calendar dates and MJDs are in Appendix A, and per-window values accompany the machine-readable table.

Target	$N_{\rm win}$	$N_{\rm EB}$	$t_{\rm on}$ (min)	$T_{\rm span}$ (h)
Proxima Cen	4	1	50.0	1.05
Barnard's Star	4	1	50.0	1.01
Wolf 359	4	1	50.0	1.10
Sirius B	12	3	17.2	0.27
G 272-61B	4	1	31.0	0.62
G 272-61A	4	1	31.0	0.62
CD-23 14742	4	1	50.0	1.10
eps Eri	4	1	42.8	0.89
tau Cet	1	1	23.2	0.47
BD+05 1668	4	1	23.7	0.46
HD 33793	1	1	14.1	0.26
Wolf 28	1	1	48.4	0.98
GJ 674	4	1	50.0	1.04
GL 3379	4	1	45.9	0.91
LSR J1835+3259	4	1	43.1	0.80
HN Lib	4	1	24.2	0.49
GJ 581	4	1	50.0	1.03
lp 876-10	4	1	46.5	0.74
61 Vir	4	1	37.8	0.95
chi01 Ori	4	1	46.4	0.92
GJ 849	4	1	44.4	0.88
gam Lep	8	2	63.7	1.23
HD 285968	4	1	46.9	0.93
AU Mic	12	3	136.1	2.93
AT Mic B	4	1	14.6	0.32
AT Mic AB	4	1	14.6	0.32
gam Vir	8	2	67.7	1.28
HR 1010	8	2	70.1	2.02
TRAPPIST-1	12	3	168.4	2.91
HD 69830	8	2	92.4	2.04
CD-38 10980	4	1	9.6	0.18
LP 349-25	4	1	44.2	0.82
GJ14	4	1	51.1	0.90
LHS 1140	4	1	43.3	0.95
HD 38858	4	1	0.3	0.03
HD 207129	4	1	47.9	1.04
HD 23484	4	1	45.9	0.91
HD 10647	8	3	107.2	1.98
g Lup	3	1	8.7	0.12
eta Crv	8	2	98.6	2.04
HD 53143	4	1	41.8	0.84
Wolf 219	4	1	1.0	0.02
bet Pic	15	3	93.7	1.81
eta Cru	4	1	30.2	0.59

($p = 0.97$); no channel within ±10 of the first-epoch flagged channel exceeds 3.1; nothing is localised to the stellar position. The second epoch's band rms (1.48 mJy per 488-kHz channel) matches the first epoch's 1.9 mJy, so the test would have recovered a comparable excess, and the frequency recurs at no other target in this release. It does not constrain intermittent or precisely-scheduled transmissions outside the sampled windows, a caveat quantified by the duty-cycle analysis of §6.

Statistical scale. The on-source peak (5.97) lies at the 92.8th percentile of the per-window control-maxima distribution of all 417 windows (median 4.55, 90th percentile 5.84; Figure 14, online-only appendix): thirty windows (7.2 per cent) have control maxima at or above it, add-one empirical $p = 0.074$, before any correction across 417 windows that would inflate it by more than two orders of magnitude: the kind of marginal excess the rate calculation predicts (one unattributed crossing among 402 non-β Pictoris windows against an expectation of 0.78, $P(\geq 1) = 54$ per cent). The candidate coincides with no catalogued molecular transition (287 MHz from CO J=2→1), and closure-phase vetting finds it consistent with a point source, one-sided evidence only (Appendix F).

TABLE 22

Masked bandwidth (GHz) by species and ALMA band on the operative mask (defined in the text). Columns B3–B8 and 'Total' are window-summed (denominator: the 700.8 GHz gross searched bandwidth); 'Union' is the same intervals merged across windows and stars (denominator: the 92.4 GHz unique-frequency union). Species rows overlap where two masks share a channel, so rows exceed the all-species totals (41.4 GHz window-summed; 4.96 GHz of the union). Per-star intervals are in the released veto-mask export.

Species	B3	B4	B5	B6	B7	B8	Total	Union
CO	0.12	0.00	0.00	1.51	0.86	0.00	2.49	0.33
^{13}CO	0.35	0.05	0.07	1.71	0.51	0.00	2.69	0.74
C^{18}O	0.04	0.00	0.00	0.00	0.00	0.00	0.04	0.04
HCN	0.00	0.00	0.00	0.00	0.12	0.00	0.12	0.12
HCO$^+$	0.10	0.00	0.07	0.00	0.36	0.00	0.53	0.50
CS	0.00	0.00	0.07	0.57	0.11	0.00	0.75	0.29
CN	0.87	0.00	0.00	11.04	2.59	0.00	14.50	2.69
SiO	0.00	0.00	0.00	0.27	0.81	0.00	1.08	0.26
All species (merged)	1.48	0.05	0.21	15.10	5.36	0.00	22.20	4.96

TABLE 23

One row per searched star, ordered by distance, in two side-by-side blocks that read down the left block and then down the right. d is the distance used by the search; "Bands" the ALMA bands in which that star has at least one searched window; $N_{\rm w}$ the number of searched windows; EIRP$_{5\sigma}$ the deepest (smallest) nominal 5σ threshold of any of those windows, in W. Star names are the identifiers of the released machine-readable table (`per_target_results_v3.28.csv`) and cross-reference it directly. 85 stars, 79 independent systems, 417 searched windows; de-duplicated, noise-defect and withheld windows are excluded here but are present, and labelled, in the released file.

Star	d (pc)	Bands	$N_{\rm w}$	EIRP$_{5\sigma}$	Star	d (pc)	Bands	$N_{\rm w}$	EIRP$_{5\sigma}$
Proxima Cen	1.30	6	4	3.30e+13	PM J034331958	20.76	6	4	1.01e+15
NAME Barnards star	1.83	6	4	6.15e+13	V star TX PsA	20.83	7	4	2.75e+14
Wolf 359	2.41	6	4	7.84e+13	V star WW PsA	20.87	7	4	2.76e+14
alf CMa B	2.67	3,4,5	12	2.34e+13	j1256-1257	21.15	7	4	8.32e+14
G 272-61B	2.67	3	4	1.62e+13	hd92945	21.51	7	4	2.65e+14
G 272-61A	2.72	3	4	1.67e+13	WD2039-202	21.77	6	4	1.29e+15
CD-23 14742	2.98	6	4	1.12e+14	HD 45184	21.89	6	4	3.03e+15
tau Cet	3.65	6	1	3.79e+13	LP 476-207 783296	23.75	7	4	4.85e+14
BD05 1668	3.79	6	4	2.71e+14	LP 476-207 384128	23.79	7	4	4.73e+14
HD 33793	3.93	6	1	5.62e+14	HD202628	23.79	6	4	1.77e+14
Wolf 28	4.31	6	1	6.67e+13	HD 31392	25.76	6	4	4.85e+15
GJ 674	4.55	6	4	2.54e+14	CD-57 1054	26.87	7	4	7.63e+14
GL 3379 Gaia DR3 3316364602541746048	5.21	6	4	1.12e+14	LP 353-51	27.10	7	4	1.23e+15
LSR J18353259	5.69	3	4	5.12e+13	hd107146	27.47	7	4	7.26e+15
HN Lib	6.25	6	4	1.49e+14	L 836-122	28.70	7	4	7.11e+14
GJ 581	6.30	6	4	6.01e+14	HD172555	28.79	6	4	1.21e+14
lp 876-10 Gaia DR3 6623351805412369024	7.68	7	4	2.70e+13	HD161868	29.75	6,7	8	1.88e+15
61 Vir	8.53	7	4	4.52e+14	51 Eri	29.91	6	4	1.09e+15
chi01 Ori	8.70	3	4	1.08e+14	HD 155555C	30.41	7	4	6.91e+14
GJ 849	8.81	6	4	8.30e+14	2MASS J05241914-1601153 551040	30.99	7	3	3.62e+15
gam Lep	8.91	6,7	8	9.27e+13	WD0346-011	31.04	6	4	2.54e+15
HD 285968	9.49	6	4	3.45e+14	2MASS J05241914-1601153 717696	31.19	7	3	3.66e+15
AU Mic	9.71	3,6	12	5.26e+13	UCAC2 20312880	33.77	7	4	1.06e+15
V AT Mic B	9.81	7	4	5.60e+13	HR 2562	33.93	7	4	3.39e+15
NAME AT Mic AB Gaia DR3 6792436799475128960	9.92	7	4	5.82e+13	TWA 7	34.10	7	4	7.39e+14
gam Vir	12.02	6,7	8	2.24e+14	WD 0407179	34.11	6	4	2.76e+15
HR 1010 Gaia DR3 4722135642226902656	12.04	6,7	8	6.50e+14	UCAC3 176-23654	34.35	7	4	9.42e+14
TRAPPIST-1	12.47	3,6,7	12	5.76e+13	GJ 2006B	34.98	7	4	8.04e+14
HD 69830	12.57	6,7	8	3.91e+13	ALMA J153702653-33192492	35.01	6,7	9	3.20e+15
CD-38 10980	12.91	6	4	4.14e+14	GJ 2006A	35.03	7	4	7.95e+14
LP 349-25	14.13	3	4	2.57e+14	TYC 2211-1309-1	35.42	7	4	9.12e+14
GJ14 Gaia DR3 383426372059603712	14.69	6	4	1.21e+14	HD14055	35.74	6,7	8	7.85e+14
LHS 1140	14.96	6	4	6.24e+14	HD 48370	35.95	6,8	8	2.86e+15
HD 38858 Gaia DR3 3023711269067191296	15.21	6	5	6.96e+14	HD 89452	36.23	6	4	8.54e+15
HD 207129 Gaia DR3 6564091190988411520	15.56	6	4	7.62e+14	HD61005	36.45	6,8	8	7.60e+14
HD 23484 Gaia DR3 4856592239127350400	16.17	6	4	1.70e+15	CP-72 2713	36.72	6,7	8	8.26e+14
HD 10647	17.35	6,7	8	1.49e+14	HD170773	36.93	6,7	8	5.47e+14
g Lup	17.40	6	3	2.55e+14	TWA 3A 696000	37.05	6	4	3.13e+14
eta Crv	18.24	7,8	8	4.36e+13	Barta 161 12	37.28	7	4	1.13e+15
HD53143 Gaia DR3 5479222240596469632	18.34	6	4	2.84e+15	HD377	38.40	6	4	1.48e+16
Wolf 219	18.88	6	4	8.43e+14	HD 139084B 921024	38.72	7	4	1.71e+15
bet Pic	19.63	3,6	15	2.34e+13	HIP 490	38.83	6	4	4.36e+14
eta Cru	19.69	6	4	1.11e+15					

Two pieces of astrophysical context are recorded for future re-examination; neither grounds the rejection. AU Mic is an exceptionally active flare star, but the candidate block's own data test that: its broadband continuum sits at a quiescent 0.15 mJy, the two flares of MacGregor et al. (2020) (16.8 and 6.1 mJy, optically thin, GHz-wide) belong to a later epoch, and an 11 mJy narrowband event over this block's quiescent level would be a factor $\sim$70 its own continuum does not show. A flare origin would need a spectrally narrow coherent process leaving the continuum untouched, which the data neither require nor establish, and the same-tuning second-epoch twin shows no burst domination. Second, the control positions sit in the same primary-beam footprint, so smooth structure such as AU Mic's debris belt raises star and controls alike and cannot by itself produce a candidate.

THE POPULATION-LEVEL CONTINUUM ANOMALY SCREEN: SUMMARY AND FULL SPECIFICATION

Because the sample is a census rather than an interest-selected list, it supports an ancillary screen: whether a star's continuum flux is anomalous against its own photospheric expectation (Planck function with Gaia DR3 parameters; Pecaut & Mamajek fallback radius where GSP-Phot is unavailable) and against same-$T_{\rm eff}$ comparison stars, in coarse bins with a minimum of five comparisons (specification and error model below). The screen is exploratory, not a headline technosignature constraint: the samples are small (57 measurements: 17 detections, 40 limits; bins of 4–7 stars) and a millimetre excess is at best an indirect, model-dependent observable of waste-heat technology, so we attach no strong technosignature interpretation to it. Its outcome this release is one outlier, χ^1 Ori at Band 3 (excess ratio 2.52, the largest of the 17 detections; resampled tail probability 0.53 in its six-star bin, the ranked outlier of a small sample and not a significant departure), consistent with astrophysical channels: it is a spectroscopic G0V binary, hence an unresolved companion or chromosphere (Appendix M).

Radius sourcing. Inputs are Gaia DR3 photometric temperatures and, where available, Gaia GSP-Phot radii. Where a direct Gaia radius is unavailable, close to half the sample by construction, the pipeline substitutes a coarse Pecaut & Mamajek main-sequence $T_{\rm eff}$–radius relation, whose scatter then propagates into the excess-ratio distribution; stars can also depart from a mean relation through age, metallicity, or unresolved multiplicity. A sensitivity check reruns the whole screen with Pecaut & Mamajek parameters imposed on every star, so the fallback's scatter can be judged against the outlier threshold: the sole flag survives, the M and G/F bin scales are unchanged (those bins are already fallback-dominated), and the A/F-hot median moves from 4.3 to 3.9.

Error model. Before any flag is evaluated the error model is $\sigma^2_{\rm tot} = \sigma^2_{\rm map} + (f_{\rm cal}\,S_\nu)^2 + \sigma^2_{\rm model}$, with $\sigma_{\rm map}$ the image rms, $f_{\rm cal}$ the band-dependent absolute-flux systematic (5 per cent in Bands 3–6, 10 per cent in Bands 7–8), and $\sigma_{\rm model}$ the photospheric-model uncertainty propagated from the Gaia temperature and radius errors, and from the main-sequence-relation scatter for the fallback stars. The per-target file's $\sigma_{\rm tot}$ column carries the map and calibration terms, $\sigma_{\rm tot} = \sqrt{\sigma^2_{\rm map} + (f_{\rm cal}S_\nu)^2}$; the model term enters the population-level excess-ratio screen, where it matters, not the tabulated per-star errors. Excess ratios near unity should be read with both the model term (dominant for fallback stars) and the calibration term (dominant for the brightest detections) in mind.

Binning and the outlier statistic. Excess ratios are compared against stars of similar effective temperature in coarse, equal-width $T_{\rm eff}$ bins, through a robust median/MAD outlier statistic. The bin width is a fixed pipeline configuration constant, published with the machine-readable release and chosen so that populated bins meet the five-comparison-star minimum. That minimum makes the screen exploratory at this depth: with five comparisons the bin median rests on two or three values, the MAD scale is itself a small-sample statistic, and repeating the comparison for every star and bin brings a small expected number of chance flags. The test is a screening tool motivating inspection of flagged stars, not a per-star hypothesis test at a fixed false-alarm rate; raising the minimum to ten comparisons, which the completed survey will reach in the occupied bins, is deferred to that survey.

Exact status in this release. The screen is live. Stellar parameters were ingested for every continuum-measured target (Gaia GSP-Phot for 16 of them, Pecaut & Mamajek otherwise), and the 17 detections among the 57 evaluated measurements carry excess ratios binned M/G/F/A-hot (7/6/4 members of the three populated bins), with per-bin median ratios and log-scatters in the machine-readable release. One measurement crosses the outlier threshold (χ^1 Ori, Band 3, $z = 4.1$); its disposition and the two committed robustness reruns (fallback-only parameters; leave-one-out bin scatter) are recorded above. The populated bins hold 4–7 detections, so every bin statistic here is small-sample.

CONTINUUM SEARCH (EXPERIMENTAL ANCILLARY ANALYSIS)

The continuum lane is an experimental, ancillary analysis rather than a co-equal search, for the reasons given in Appendix L.

Of the 57 target/bands with a usable continuum measurement, seventeen show a $\geq 5\sigma$ detection and forty are non-detections (Figure 6, right panel); a continuum "upper limit" is 5× the map rms at the star's position, primary-beam corrected as the EIRP thresholds are, and the per-target rms values of Appendix I rescale it to any other confidence level. The non-detections reach 0.036–3.6 mJy (median 0.52 mJy), the deepest towards TRAPPIST-1 in Band 3. Under the combined error model of Appendix L, sixteen of the seventeen detections remain above 5σ; the exception (η Corvi Band 7, 4.7σ combined) is treated as marginal. Several fields reach bright peaks well away from the target (ϵ Eri Band 6 at 10.2σ, 18 arcsec from the phase centre; GJ 674 and CD−23 14742 at 10.7 and 9.9σ), none within the 3-arcsec positional tolerance of its star, and all are correctly recorded as non-detections; whether the ϵ Eri peak is real off-target emission or a primary-beam artefact at $\sim$0.7 FWHM offset is undecidable from the retained products (holography-beam recomputation committed, §4).

Every detection is astrophysically unsurprising. The G 272-61A/B pair is the unresolved UV Ceti binary (0.85 mJy, attributable to neither component). Of the rest, six are young, active systems with known debris discs or chromospheric activity (τ Cet, β Pic B3/B6, AU Mic, AT Mic A/B); six show the photospheric flux expected of their spectral types (γ Lep and γ Vir in both bands, χ^1 Ori, η Cru; η Corvi's Band 8 limit is likewise photosphere-consistent); and two are late-M dwarfs of documented strong magnetic activity, LP 349−25 (10.5σ, 0.071 mJy) and the auroral emitter LSR J1835+3259 (6.7σ, 0.078 mJy; Hallinan et al. 2007), which merits deeper follow-up. Closure phase cannot discriminate structure for any of the seventeen at these flux densities (Appendix F), and the population-level anomaly screen returns one outlier among them, χ^1 Ori Band 3, with an astrophysical disposition (Appendix L).

A SERENDIPITOUS MOLECULAR-LINE CATALOGUE

The line-exclusion step of the continuum search (§4) identifies every $\geq 5\sigma$ outlier channel in each target's spectrum as a byproduct; no existing mm/submm line survey of the solar neighbourhood is drawn from a distance-limited census of archival coverage, so even a modest catalogue adds value independent of its SETI motivation. Across the subset scanned to date, 18 outlier channel groups were found. Twelve are single- or few-channel features with no known transition within 300 MHz, most plausibly statistical. The remaining six are plausibly real: CO(J=1→0) and CO(J=2→1) towards β Pictoris (the stellar-frame component of Table 6 and §5.4); a marginal CO(J=2→1)-consistent feature recurring across two epochs towards HD 285968, tentative only (1–2 channels of significance per epoch); and a single-channel, 195-MHz-offset feature near SiO(J=5→4) towards HD 53143, more likely statistical than real. All are reported openly, including the two we consider probably spurious; the machine-readable catalogue (per-group target, band, channel coordinates, significance, nearest transition) ships with the tagged public release, and the completed survey's full-sample re-scan will extend it.

LESSONS FOR ARCHIVAL PIPELINES

Three of the four audit items of §5.3 are failure modes generic to any survey mining a heterogeneous archive with automated crossmatch and processing pipelines. First, every processing lane needs its own geometric guard: the narrowband step aborts on the 52-arcmin pointing mismatch of item (ii) while the continuum-imaging step had none, an asymmetry by omission rather than necessity, which the corrected release carries on both lanes. Second, window-level bookkeeping anomalies are caught by cross-target pattern recognition, not per-target sanity checks: the defect of item (iii) recurred with matching correlator parameters across two unrelated targets, and it was the recurrence, not either instance, that established a pipeline artifact rather than a target idiosyncrasy. Third, name resolution is not position resolution. The γ Lupi entry was misidentified through the SIMBAD identifier list, not any positional error: HD 139664 itself carries the historical main identifier "* g Lup", so a name-based lookup of "γ Lupi" returns it rather than the B2IV star seven times farther away (parallax 7.75 mas, 3.7° distant on the sky). What caught it was cheap and should be standing practice: the observed star's Gaia DR3 parallax, 57.476 mas, recovered digit-for-digit by inverting the pipeline distance, with $T_{\rm eff} \approx 6600$ K and cross-identifiers HR 5825, HIP 76829 and GJ 594 consistent with F4V. An archival survey inherits the catalogues' historical artefacts; position-anchored verification is the inexpensive antidote.

FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

Versions up to 3.24 carried a four-model bookkeeping ladder (execution-block permutation, hypergeometric, block bootstrap, and a Poisson rate heuristic, spanning $P(F \geq 3) = 6.1$–9.2 per cent) whose sole purpose was to calibrate a false-alarm rate for an *asymmetric* statistic whose per-window null rate could not be written down. The symmetric statistic of §5.4 removes that need, giving a per-window rate of 1/513 by construction and a survey-level expectation of $417/513 = 0.81$ flags. The ladder is therefore retired rather than updated, and this appendix records what replaces it and what the replacement cannot do.

Two limits bound the replacement. The 1/513 rank floor of §5.4 can be lowered only by enlarging the control ring, which the primary beam bounds; and windows are not fully independent, being grouped into 100 execution blocks sharing weather, calibration, and correlator setup, so $N/513$ is an expectation over a mildly correlated sample rather than N independent trials. Both push conservatively for the conclusion actually drawn, that the flagged windows are consistent with chance, because they widen rather than narrow the plausible null range.

THE SYMMETRIC STAR-VERSUS-CONTROL NULL IN FULL

Two details of the test of §5.4 belong here. Its per-window p-value is $p = (1 + \#\{T_i \geq T_\star\})/(N+1)$; and its seven measured windows are the six validation windows of the local reprocessing set plus the AU Mic candidate's first-epoch window, reprocessed from the raw execution block with the observatory pipeline's own calibration recipe (§5.5). It is the designated replacement for the empirical star-versus-ring calibration as the remaining measurement sets are re-run. The AU Mic candidate also fails the single-position variant of the release statistic (star 5.22 against its ring maximum 5.85, §5.5).

Table 16 consolidates the family-wise trials factors: every per-window factor is applied identically at the 512 control positions, which makes the control ring, and the empirical star-beats-ring rate built on it, the operative calibration rather than any analytic trials correction.

TABLE 24
Exceptions of the integration-time audit (§5.6). The 53 remaining star–execution-block groups all have pipeline on-source time at or below the archive science exposure, so the comparison is conservative for them; three further groups (Wolf 28, the narrow AT Mic window, and G 272-61) agree to better than 1%.

Star / block	Pipeline (s)	Archive (s)	Ratio
AT Mic (A002_Xcd97a1_X3f1e, wide windows)	1258	877	1.43
CD−38 10980 (A002_Xe5808e_X80a1)	617	575	1.07

ROBUSTNESS OF THE EMPIRICAL CALIBRATION

The chance-crossing expectation is also consistent with the trials count once adjacency is handled. The release evaluates 3.97×10^7 channel×drift trial cells; a Gaussian one-sided 5σ tail (2.87×10^{-7}) over that count predicts 11.4 chance crossings, against the 18 windows containing at least one on-star crossing. Raw counts run higher because the drift-tracked statistic is strongly correlated between adjacent channels ($\rho_{\mathrm{lag1}} = 0.989$–$0.991$, i.e. 8.5×10^4–1.1×10^5 effectively independent channel cells rather than 3.97×10^7), so crossings arrive in contiguous runs. Collapsed to clusters (kernel of 3 native channels) they leave far fewer independent events, and it is that cluster count, not the raw count, that must be compared with the trials expectation; the two agree. The per-crossing census for the extended sample is in the released products.

The star-beats-ring rate is not driven by any single stratum. The stratification below was executed on the $\leq$20 pc subset, whose counts it quotes. By band: 20 per cent (B3, $n = 40$), 10.9 per cent (B6, $n = 128$), 17.8 per cent (B7, $n = 45$), zero in the sparsely sampled B4/B5/B8 ($n = 4$ each). By channelisation: 17.0 per cent for fine windows ($n = 47$) against 12.4 per cent for coarse ($n = 178$). By primary-beam offset: 15.7 per cent above the median offset (0.10 arcsec) against 11.0 per cent below. The disc-hosting caveat is empirically inverted: the 132 windows towards disc- or planet-hosting stars contribute all eight crossing windows and all three candidates but only 13 star-beats-ring wins (9.8 per cent), while the 93 towards non-hosts contribute no crossings yet 17 wins (18.3 per cent), so real circumstellar emission shared by star and ring is not what drives the statistic. Excluding the fifteen β Pictoris windows leaves one unattributed crossing among 402 windows, against 0.78 expected ($P(\geq 1) = 54$ per cent) on the empirical calibration.

The control ensemble's geometry carries a second property beyond the shared primary-beam gain of §5.4: neighbouring positions are correlated below the synthesised-beam scale, so the independent spatial trials fall from 512 to $N_{\mathrm{eff}} \simeq 30$–$43$, as the symmetric reprocessing runs measure directly – a floor to which the empirical 15.3 per cent rate, measured on the correlated ensembles themselves, is immune by construction. A continuum source in the ring raises the narrowband statistic only if it also emits a narrow line in-window, and the cross-target occupancy screen (Appendix H) covers recurring frequencies.

INTEGRATION-TIME AUDIT: METHOD AND EXCEPTIONS

The archive cross-check of §5.6 uses the ALMA Science Archive's `ivoa.obscore` service (queried 2026 September 8), whose rows give, per observing unit set and pointed science source, the science-intent exposure `t_exptime` aggregated over all executions of that unit set. For each of the 57 searched execution blocks the pointed science source came from the archive's field list, aliases resolved by hand (Sirius B enters through the Sirius A pointings, the UV Ceti and G 272-61 pairs share one field each, AT Mic's block points at the pair), and the pipeline's on-source time was compared with the archive's exposure for that source and unit set. Two `obscore` properties bound the comparison: it counts science-intent scans only, so target time in calibrator-intent scans is invisible to it, making the archive figure a lower bound and any pipeline over-count conservative; and per-execution durations are not exposed, so a multi-execution unit set is compared through its aggregate.

The audit covers all 227 window-level records in 58 star–execution-block groups of that subset: the 59 pairs of the level-1 bookkeeping, with the G 272-61 and AT Mic binaries each collapsed to their shared pointing and AT Mic's narrow flagband window carried separately from its wide windows. It is conservative for every window of the two candidate hosts of §5.4, since β Pictoris's windows (1824–1845 s) sit inside an archive science exposure of 47 444 s and AU Mic's inside 109 334 s: no candidate disposition changes, and the archive's values lower the 1.4×10^6 s GHz exposure-weighted coverage total of §5.4 by less than 1 per cent.

The AT Mic block carries the diagnostic signature of a dump-counting defect: three of its windows report 208 integrations where 145 exist, and 1258 s of on-source time against the block's own 1140 s elapsed span. This paper's duty table uses the archive-consistent 877 s for both AT Mic catalogue entries; the over-counted values survive only in per-window machine-readable fields, to be corrected in the next release's export. The CD−38 10980 excess (7%) is smaller than the intent-filter's conservative margin and its limit is unaffected (thresholds derive from measured rms, not integration time). A Wolf 359 discrepancy in the raw comparison is an audit-side artefact, an over-aggressive calibrator-source exclusion in the query itself; it disappears once the star's science-source exposure (9463 s, ample against the 2999 s searched) is included, exactly the kind of error the audit is designed to catch.

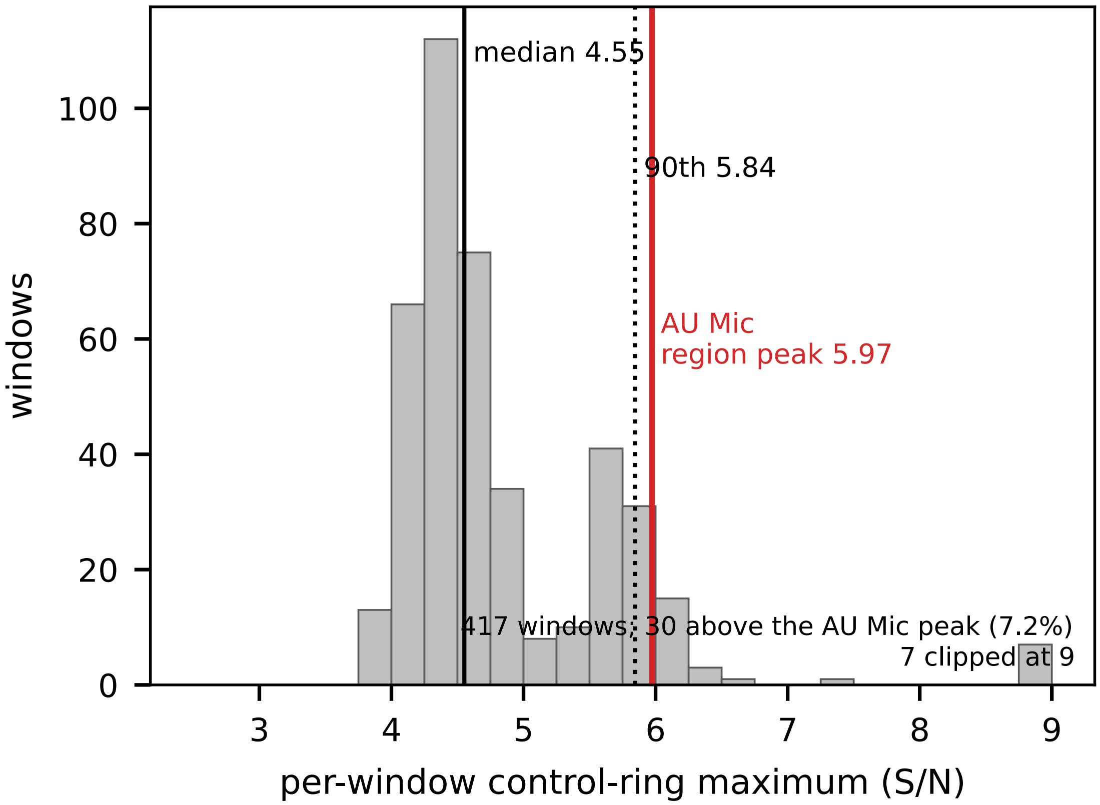

Fig. 14.— Diagnostic distribution for the marginal AU Mic candidate: the per-window control-ring peak signal-to-noise ratios of all 417 windows of this release (histogram; median 4.55, 90th percentile 5.84), i.e. the maxima expected from noise plus background structure alone. The dashed line marks the formal 5σ crossing threshold, the solid line the on-source peak of the AU Mic candidate window (5.97, the 92.8th percentile of this distribution) and the dotted line its own control maximum (5.85). Three windows lie beyond the plotted range, marked at upper right: HD 285968 Band 6 at 10.9, and β Pictoris Band 6 at 9.1 and at 15.2, the latter the CO-filled control ring discussed in the text.

EXCLUSION AND DEFECT FORENSICS

THE FOUR CLASSES OF EXCLUDED ENTRY, ITEMISED

The two process-level failures of §5.3 are item (ii) below. The six windows failing the noise-defect test lie across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice). The supporting audit is clean across all 417 windows bar the four χ^1 Ori Band 3 windows (elevated system temperature; limits weakened, not invalidated), and all 168 census identifiers re-resolve through SIMBAD and *Gaia* to 168 distinct sources inside 40 pc. The four classes of entry absent from the per-target file are itemised below, so their absence is auditable.

(i) Narrowband process failures. Four target/bands (Barnard's Star [B7], Wolf 359 [B7], SCR J1845−6357, Wolf 358) have no narrowband result after process-level search failures; each retains a valid continuum non-detection and carries no EIRP threshold.

(ii) Giclas 9−38 pair. Giclas 9−38A and Giclas 9−38B [B3] are excluded entirely: the pair's only matched member observation unit set points ∼52 arcmin from either star, 35× its own primary-beam FWHM. The narrowband step correctly aborts; the continuum-imaging step lacks the equivalent guard (Appendix O), so the target/band is excluded pending one.

(iii) Window-level noise defect. One spectral window in each of two Band 6 targets with similar correlator setups (HD 10647, HD 139664) reports 12.1 s of on-source integration against 393–520 s for its siblings in the same execution block, yet a combined noise estimate ∼1000× deeper than those siblings and than the survey's best continuum limit – the opposite of what radiometer-noise scaling predicts for a ∼40× shorter integration. Both the 12.1 s and the noise estimate come from the same machine-written per-window product that carries every other window's values, and are defective together. The affected window's narrowband result is excluded and the combined continuum measurement withheld for both target/bands, since it is built from all configured windows together; their other narrowband windows are retained.

(iv) The "γ Lupi" resolution. The object observed under the catalogue entry "γ Lupi" is the F4V debris-disc host HD 139664 at 17.40 pc (Gaia DR3 5989102478619519616), not the naked-eye γ Lupi (Appendix O). All such entries are relabelled; no measured value changes and the defect status of item (iii) is unaffected.

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